Verified Generators and Relations for Qupit Clifford Operators

XIAONING BIAN, Tsinghua University, China
YUAN FENG, Tsinghua University, China
SARAH MENG LI, University of Amsterdam and QuSoft, The Netherlands
JOHN VAN DE WETERING, University of Amsterdam and QuSoft, The Netherlands

A recent paper by Bian, Li, Ross, van de Wetering, and Zhao gives a complete set of sixteen rewrite rules for multi-qudit Clifford circuits in every odd prime dimension p: two circuits over the gates H, S, and CZ implement the same unitary exactly when the rules prove them equal. The completeness proof is a normal-form argument in the tradition of Selinger's presentation of the qubit Clifford group; its heart is a case analysis pushing every generator through every normal box, and its supporting derivations run to roughly a hundred pages. This paper reports a machine-checked proof of the result, in safe, axiom-free Agda, against an independently constructed semantics. The formalisation does not follow the paper's proof. The paper works with one rule set for the whole Clifford group and applies it, implicitly, "up to Paulis"; the step that promotes a symplectically complete rule set to a projectively complete one (its Proposition 4.9) is convincing on paper but hard to turn into a verified normalisation procedure. We instead factor the projective Clifford group as a semidirect product $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$ of the projective Pauli group by the symplectic group, present the two factors separately, and glue the presentations with a generic, verified semidirect-product theorem. The symplectic factor carries all the mathematics: a unique normal form for $\mathrm{Sp}(2n, \mathbb{Z}_p)$, refined here into a doubly inductive datatype of *normal boxes* and *rows*, is realised as a tower of verified coset tables in the style of Reidemeister and Schreier, its uniqueness is proved by induction on the width against the symplectic action on Pauli vectors, and the paper's 66 box relations, which the rewriting uses, are derived from eighteen natural ones. The projective Pauli factor is assembled compositionally. Presentation-level "plumbing" then reduces the glued alphabet and relations to the paper's own generators and rules, and a verified isomorphism of presentations connects the two routes: the paper's Figure 1 read modulo scalars and the semidirect presentation present the same group. Finally, the scalars are restored by a central extension, verified against a cocycle construction. Everything is parametric in the prime p and typechecks with no postulates; the Agda development is about 100 000 lines, of which 60 000 are specific to the qupit Clifford groups.

CCS Concepts: • **Theory of computation** → **Type theory**; **Logic and verification**; **Equational logic and rewriting**; **Quantum computation theory**.

Additional Key Words and Phrases: Clifford group, qudits, generators and relations, completeness, normal forms, symplectic group, semidirect product, Agda, formal verification

ACM Reference Format:
Xiaoning Bian, Yuan Feng, Sarah Meng Li, and John van de Wetering. 2026. Verified Generators and Relations for Qupit Clifford Operators. *Proc. ACM Program. Lang.* 1, OOPSLA, Article 1 (October 2026), 33 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

A *presentation by generators and relations* of a group of quantum operators is a finite list of circuit equations that is *sound* — each equation holds in the intended semantics — and *complete* — every semantic equality between circuits is derivable from the list. Soundness is a finite collection of

Authors' Contact Information: Xiaoning Bian, Tsinghua University, Beijing, China, bian@dal.ca; Yuan Feng, Tsinghua University, Beijing, China, yuan_feng@tsinghua.edu.cn; Sarah Meng Li, University of Amsterdam and QuSoft, Amsterdam, The Netherlands, sarah.li@uva.nl; John van de Wetering, University of Amsterdam and QuSoft, Amsterdam, The Netherlands, john@vdwetering.name.

2026. ACM 2475-1421/2026/10-ART1
https://doi.org/10.1145/nnnnnnn.nnnnnnn

matrix identities. Completeness is a different animal: the standard proof defines a *normal form* for circuits, shows that every circuit rewrites to a normal form using the relations, and shows that distinct normal forms denote distinct operators. The middle step is a case analysis — every generator against every piece of every normal form — that is mechanical, unforgiving and enormous. Selinger's presentation of the n-qubit Clifford group [Selinger 2015] runs a substantial rewrite system through such an analysis; its qutrit successor [Li et al. 2025], the first completeness result for circuits in an odd prime dimension, needs more than fifty pages; and its generalisation to all odd primes by Bian, Li, Ross, van de Wetering, and Zhao [Bian et al. 2026a] — sixteen rewrite rules for multi-qudit Clifford circuits in every odd prime dimension p, henceforth "the qupit paper" — rests on roughly a hundred pages of derivations spread over its appendices and a supplement [Bian et al. 2026b]: sixty-six box relations, the conjugation tables of every normal box, the pushing lemmas, and the reductions of the box relations to eighteen symplectic relations and of those to the final sixteen. Proofs of this shape — long, regular and computational — are exactly the proofs that a proof assistant with a strong evaluator can check largely *by computation*, and exactly the proofs whose hand-checked versions one would like to see machine-checked.

This paper reports such a machine-checked proof. We formalise, in safe, axiom-free Agda [Norell 2007; The Agda Community 2026], that the qupit paper's rule set presents the multi-qupit Clifford group, at every width n and every odd prime p, against a semantics constructed independently of the syntax. The formalisation is built on a library for presentations, normal forms and completeness of circuit relation sets that the authors developed for this purpose [Bian and Feng 2026a]; the library's design — inductively defined circuits, presented monoids as inductive congruences, normal forms as function bundles, a verified Reidemeister–Schreier engine, and presentation theorems for products and extensions — is described in a companion paper [Bian and Feng 2026b], and we recall the parts we use in §3.

A different proof. The formalisation does *not* follow the qupit paper's proof, and the reasons are instructive. The paper works with a single rule set for the whole Clifford group and organises its completeness proof around the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_p)$, the Clifford group modulo Paulis: it shows that every circuit rewrites, *up to Paulis*, to a symplectic normal form, and then boosts symplectic completeness to projective and unitary completeness by a general argument (its Proposition 4.9) in which each rewrite is repaired by pushing a Pauli correction to the end of the circuit. Throughout, relations are used "projectively": Paulis and scalars are silently dropped and later restored. This is sound mathematics, and we believe it; but it is a poor specification for a verified normalisation procedure. The rewriting steps do not preserve any one syntactic invariant, the correction Paulis are not part of the data the rewrites compute with, and the termination and well-definedness arguments (its Lemma 4.2 and Proposition 4.9) are stated at a level of abstraction — "push the Pauli to the right, it changes to some other Pauli" — that a proof assistant would have to reconstruct from scratch.

We therefore chose a route that a proof assistant likes. The projective Clifford group is a *semidirect product* of the projective Pauli group by the symplectic group, and the Clifford group is a *central extension* of that by the scalars. We present the three factors separately — the symplectic group by a coset-table normal form, the Pauli group compositionally, the scalars as a cyclic group — and glue the presentations together with generic, verified presentation theorems for semidirect products and central extensions. On this route the Paulis are never "implicit": the symplectic development never mentions them, and they enter only through the conjugation action of the semidirect product, which is verified once, by semantics, rather than threaded through every rewrite. A final layer of presentation-level "plumbing" reduces the glued alphabet and relations to the paper's generators

H, S, CZ and its rules, and a verified isomorphism of presentations connects the two routes: the paper's Figure 1 read modulo scalars presents the same group as our semidirect presentation.

1.1 Contributions

(1) **A machine-checked presentation of the projective qupit Clifford groups.** For every odd prime p and every width n, the paper's Figure 1 read modulo scalars presents $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$ (Theorem 2.1, §2), via a chain of verified isomorphisms of presentations from a semidirect presentation on the alphabet X, Z, H, S, CZ down to the paper's fifteen rules on H, S, CZ (§6).

(2) **A machine-checked presentation of** $\mathrm{Sp}(2n, \mathbb{Z}_p)$ **and a verified unique normal form.** Eighteen relations (fifteen group-specific and three structural) present the symplectic groups; the normal form is the paper's symplectic normal form refined into a doubly inductive datatype of *normal boxes* and *rows*, realised as a Reidemeister–Schreier coset tower, and proved unique against the symplectic action on Pauli vectors (Theorem 2.2, §5).

(3) **The relation reductions, verified.** The box relations that the normalisation uses are derived from the full symplectic rules, those from the simplified ones, the semidirect relations from the Clifford ones, and the paper's rules from ours; each reduction is an isomorphism of presentations and each is checked (§5.7, §6.3).

(4) **The scalars restored.** The exact rule set — the sixteen rules corrected by the phase each actually picks up, which is a single power of ω on one rule — presents a central extension of the projective group by $\langle\omega\rangle$, verified against the Weyl cocycle $\frac{1}{2}$ sform (§6.4).

(5) **A methodological comparison.** We explain precisely where the paper's proof resists formalisation and how the divide-and-conquer route avoids each difficulty (§4), and we report what the formalisation covers and what it does not (§7).

Everything stated as a theorem in this paper typechecks under `-safe`: there are no postulates, no holes and no classical axioms. The two facts that are *not* formalised are the glue between our constructed groups and the physicist's: that the projective Clifford group of unitaries is $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$, and that the Clifford group is a central extension of it by $\langle\omega\rangle$. Both are classical [Bian et al. 2026a], and neither involves circuits; §7 states exactly what our semantic groups are.

1.2 Outline

Section 2 recalls qupit Clifford operators, states the paper's theorem, and states our two main theorems as Agda types. Section 3 recalls the framework. Section 4 contrasts the two proofs. Sections 5 and 6 are the method: the symplectic presentation, then the composition of the three factors. Section 7 gives statistics, engineering lessons and caveats; §8 surveys related work; §9 concludes.

2 Qupit Clifford Circuits and the Main Theorems

2.1 From qubit to qudit Clifford gates

Fix an odd prime p and let $\omega = e^{2\pi i/p}$. A *qupit* is a p-dimensional quantum system with computational basis $|0\rangle, \ldots, |p-1\rangle$, indices read in $\mathbb{Z}_p$. The single-qupit Pauli operators are $X|j\rangle = |j+1\rangle$ and $Z|j\rangle = \omega^j |j\rangle$; the n-qupit Pauli group $\mathcal{P}_n$ is generated by tensor products of X and Z together with the phases ω^t. The n-qupit Clifford group C_n is generated, under composition and tensor product, by the gates of Table 1, whose qubit counterparts are shown for comparison. We follow the qupit paper's conventions [Bian et al. 2026a]: the $\frac{1}{2}$ in S is the inverse of 2 in $\mathbb{Z}_p$, and H carries a global phase δ_p chosen so that all three generators have determinant one (which is what makes the Clifford phases exactly the powers of ω).

Table 1. The Clifford generators for qubits and for qupits (odd prime p).

	qubit ($p = 2$)	qupit (odd prime p)
H	$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$	$\lvert j\rangle \mapsto \frac{\delta_p}{\sqrt{p}} \sum_k \omega^{jk} \lvert k\rangle$
S	$\begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}$	$\lvert j\rangle \mapsto \omega^{j(j-1)/2}\,\lvert j\rangle$
CZ	$\mathrm{diag}(1,1,1,-1)$	$\lvert j\rangle\lvert l\rangle \mapsto \omega^{jl}\,\lvert j\rangle\lvert l\rangle$

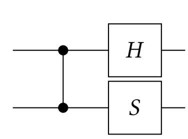
Three quotients of C_n matter. Writing $\langle\omega\rangle$ for the scalars, $\overline{C}_n = C_n/\langle\omega\rangle$ is the *projective* Clifford group and $\overline{\mathcal{P}}_n = \mathcal{P}_n/\langle\omega\rangle \cong \mathbb{Z}_p^{2n}$ the projective Pauli group, an abelian group in which X and Z commute. Conjugation by a Clifford operator permutes Paulis and preserves their commutation phases, so modulo Paulis a Clifford operator is a symplectic matrix: $C_n/\mathcal{P}_n \cong \overline{C}_n/\overline{\mathcal{P}}_n \cong \mathrm{Sp}(2n, \mathbb{Z}_p)$, the symplectic group of the $2n$-dimensional $\mathbb{Z}_p$-vector space $\mathbb{Z}_p^{2n}$ with its standard symplectic form [Bian et al. 2026a]. Bringing the Paulis to the other side, the projective Clifford group splits: $\overline{C}_n \cong \overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$, the semidirect product with $\mathrm{Sp}(2n, \mathbb{Z}_p)$ acting on $\overline{\mathcal{P}}_n$ by its defining action on $\mathbb{Z}_p^{2n}$. These are the “glue facts” of the introduction; our formal development *defines* its semantic groups as the right-hand sides.

2.2 Circuits in Agda

In the formalisation a gate set is a family of gate types indexed by arity, and a circuit on n wires is a *word* over wire-indexed generators (§3 gives the generic machinery):

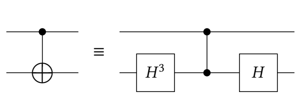
```
data SympGate : ℕ → Set where
  H-gate   : SympGate 1
  S-gate   : SympGate 1
  CZ-gate  : SympGate 2

example : Circuit 2
example = CZ • H ↑ • S
```

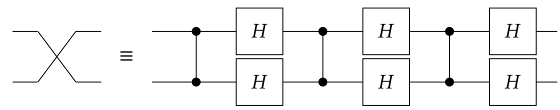
`Circuit n = Word (Gen n)` is the free monoid on generators; `_↑` shifts a circuit up one wire.

Diagrams are read in matrix-multiplication order, so `example` is 

. The remaining usual gates are *words*, not extra generators — X, Z, the swap Ex and the two controlled-X gates:

```
Z X : ∀ {n} → Word (Gen (1+ n))
Z = H • H • S • H • H • S⁻¹
X = H • S • H • H • S⁻¹ • H

Ex CX XC : ∀ {n} → Word (Gen (2+ n))
Ex = CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑
CX = H ↓ ^ 3 • CZ • H ↓
XC = H ↑ ^ 3 • CZ • H ↑
```





Here $S^{-1} = S^{p-1}$, $H^{-1} = H^3$ and $CZ^{-1} = CZ^{p-1}$ are the words that the relations make inverses; _↓ is the identity on circuits and exists to make "the lower copy of a gate" visible in a rule. One more derived gate has no qubit counterpart: the *multiplier* $M_x|j\rangle = |xj\rangle$ for a unit $x \in \mathbb{Z}_p^*$. The qupit paper spells it (its Lemma 2.14) as $HS^{g^{-1}}HS^{g}HS^{g^{-1}}$ followed by a Pauli correction and a phase; the QPL talk on the paper spells it as $R^x\, H\, R^{x^{-1}}\, H\, R^x H$ with $R = S \cdot Z^{1/2}$, which absorbs the Pauli. We shall meet several spellings, one per rule set, and one of the results below is that they are all interderivable.

2.3 The whole development is one parameterised module

Everything below is parameterised by the prime and by a *multiplicative generator* g of $\mathbb{Z}_p^*$:

```
module Qupit-Clifford-Theorems
  (p-3 : ℕ)
  (let p-2 = 1+ p-3)
  (p-prime : Prime (2+ p-2))
  (let open PrimeModulus' p-2 p-prime)
  (g*@(g , g≠0) : ℤ* p)
  (g-gen : ∀ ((x , _) : ℤ* p) → ∃ λ (k : ℤ p-1) → x ≡ g ^ toℕ k)
  where
```

So p = p-3 + 3 ranges over every odd prime, uniformly, and the hypothesis g-gen says that every unit is a power of g, i.e. g is a primitive root. The existence of a primitive root modulo every prime is classical; we take it as a parameter rather than proving it, exactly as the qupit paper fixes a generator g of $\mathbb{Z}_p^*$ in its Figure 1. Every theorem is stated for the multiplicative generator chosen.

2.4 Main result 1: a presentation of the projective qupit Clifford group

The rule set of the QPL 2026 talk on the qupit paper — Figure 1 read modulo scalars, with the multiplier spelled over R and with $(SH)^3 = \varepsilon$ kept as an axiom — is, in Agda, one inductive family with sixteen constructors, one per rule, each stated at the smallest number of wires at which it makes sense:

```
data _CRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-S      :         (1+ n) CRel,  S ^ p === ε
  order-H      :         (1+ n) CRel,  H ^ 2 === M-1
  M-power      : ∀ k →   (1+ n) CRel,  Mg^ k === M (g^ k)
  semi-MR      :         (1+ n) CRel,  R • Mg === Mg • R^ (g * g)
  order-SH     :         (1+ n) CRel,  (S • H) ^ 3 === ε
  comm-HHSHHS  :         (1+ n) CRel,  H • H • S • H • H • S === S • H • H • S • H • H

  order-CZ     :         (2+ n) CRel,  CZ ^ p === ε
  order-Ex     :         (2+ n) CRel,  Ex ^ 2 === ε
  comm-CZ-S↑   :         (2+ n) CRel,  CZ • S ↑ === S ↑ • CZ
  semi-M↑CZ    :         (2+ n) CRel,  CZ • Mg ↑ === Mg ↑ • CZ^ g
  semi-Ex-S↑   :         (2+ n) CRel,  Ex • S ↑ === S ↓ • Ex
  semi-Ex-H↑   :         (2+ n) CRel,  Ex • H ↑ === H ↓ • Ex
```

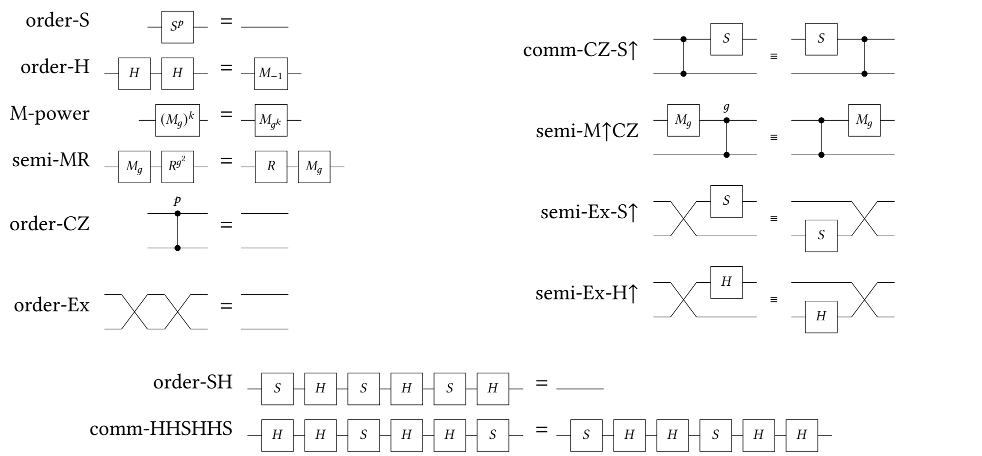


Fig. 1. The one- and two-qupit rules of _CRel,_===_ as circuit equations. Diagrams read in matrix-multiplication order, so a word $w \bullet v$ is drawn with v on the left.

```
blake-c12    :       (2+ n) CRel,
                       (S ^ p-1) ↑ • (S ^ p-1) ↓ • CX ^ p-1 • S ↓ • CX === CZ

yang-baxter  :       (3+ n) CRel,  Ex ↑ • Ex ↓ • Ex ↑ === Ex ↓ • Ex ↑ • Ex ↓
cz-slide     :       (3+ n) CRel,  Ex ↓ • Ex ↑ • CZ === CZ ↑ • Ex ↓ • Ex ↑
semi-CX↑-CZ↓ :       (3+ n) CRel,  CZ ↓ • CX ↑ === CZ02 • CX ↑ • CZ ↓
```

(In the source the multiplier is `XMg`, the “X-filling” $R^{x^{-1}}HR^{x}HR^{x^{-1}}H$ of the shape $R^{a}HR^{b}HR^{a}H$; we write `Mg` here for readability, as the talk did.) Figures 1 and 2 show the same sixteen rules as circuit equations, drawn from the Agda terms. Beyond these, *every* circuit theory in the framework gets the same three structural rules for free: congruence under ↑, gates on disjoint wires commute, and (when there is one) a global scalar gate commutes with everything. Nothing else is an axiom: $XZ = ZX$ modulo scalars, the commutation of Ex with CZ, and both rules for pushing a Pauli through CZ are *derived*. The count is sixteen where the paper's Figure 1 has fifteen non-scalar rules because the talk kept $(SH)^3 = \varepsilon$ as an axiom under the R-spelling; §6.3 shows that under the paper's own S-spelling it is a theorem, and verifies the paper's fifteen.

The first main theorem says that these rules *present* the projective Clifford group, at every width:

Theorem 2.1 (`clifford-presentation`). *For every width n, the sixteen rules present the projective Clifford group:*

```
Theorem-1 : ∀ (n : ℕ) → (n CRel,_===_) IsPresentationOf (Pauli⋊Sp n)
Theorem-1 = PapPres.presentation
```

Home: `ProjectiveClifford.Qupit.Paper-V0.Presentation` *and* `Paper-V1.Presentation`.

Here `Pauli⋊Sp` n is the semidirect product of the projective Pauli group $\mathbb{Z}_p^{2n}$ by the symplectic group, constructed in §6.2, and `_IsPresentationOf_` is the record of §3.1: a proof that every generator has an inverse, an interpretation [[_]] of words in the group, and a *group isomorphism* from the words modulo the rules onto the group. Its two halves are the two directions of the diagram

$$\mathrm{Cir}\ n \,/ \approx_{16\text{ rules}} \xrightarrow{[\![\_]\!]\ \cong} \overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p) \;\cong\; C_n/\langle\omega\rangle :$$

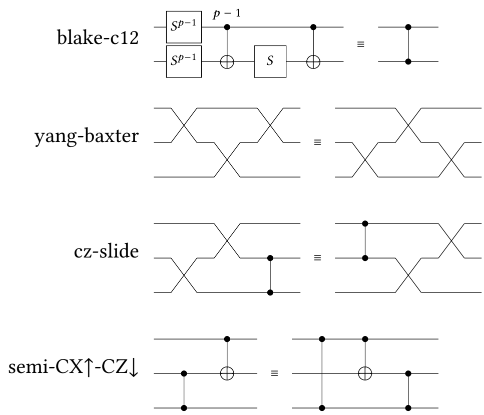


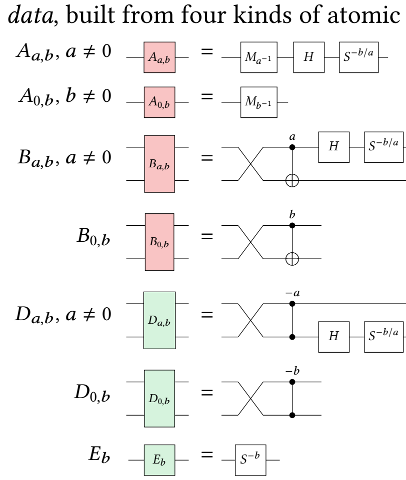
Fig. 2. The remaining two- and three-qupit rules. `blake-c12` relates two circuits implementing controlled-Z; `yang-baxter` and `cz-slide` are the swap calculus.

soundness — ⟦_⟧ is well defined, each rule holds semantically — and *completeness* — ⟦_⟧ is injective, semantically equal circuits are related by the rules. The same theorem is proved for the paper's fifteen rules in their own spelling (§6.3) and, with the scalar ω and its one correction added, for the exact Clifford group (§6.4).

2.5 Main result 2: a unique normal form for symplectic operators

Symplectic operators are Clifford operators modulo Paulis — the "proper" part of a Clifford operator, its stabiliser tableau. Our normal forms for them are *data*, built from four kinds of atomic boxes:

```
A B D E : Set
A = Σ[ ab ∈ (ℤ p × ℤ p) ] (ab ≢ (0 , 0))
B = ℤ p × ℤ p
D = ℤ p × ℤ p
E = ℤ p
```



The boxes are the paper's normal boxes (its Figure 6), with the same names; the pictures show how each is spelled as a circuit. Rows of boxes and the staircase of rows are two more inductive definitions (§5.3), and a normal form is read back as a circuit by structural recursion. The second main theorem says the normal form is unique for the symplectic semantics of §5:

Theorem 2.2 (`unique-nf`). *Normal forms with equal symplectic denotations are equal:*

```
Theorem-2 : ∀ n {u v : NF n} → ⟦ [ u ] ⟧ ≈ˢ ⟦ [ v ] ⟧ → u ≡ v
Theorem-2 = U.⟦[]⟧-injective
```

Home: `Symplectic.Normalization.Uniqueness`.

Soundness plus a unique normal form yields completeness, by one generic lemma (§3.2); and the hard, interesting part of the whole development is to find these coset representatives, parameterised by the number of wires, and to push every generator through them (§5).

3 The Framework

The formalisation is built on an Agda library for presentations of monoids and groups by generators and relations, normal forms, and completeness of finite relation sets, with wire-indexed circuits as its motivating instance [Bian and Feng 2026a,b]. This section recalls, briefly and with code quoted verbatim, the parts the rest of the paper uses. The emphasis is on two things: the generic lemma `by-normalization`, which is the only place completeness is ever proved, and the *presentation construction theorems* — for semidirect products and for central extensions — which are what allow §§5–6 to deal with presentations and normal forms of the symplectic group, the Pauli group and the scalars one at a time.

3.1 Words and presented monoids

A word over a generator set X is an unquotiented syntax tree with constructors $[\_]^{\mathrm{w}}$, ε and $\bullet$ — deliberately not a list, so that a word remembers its bracketing and rules may be applied to any parenthesised subterm. A relation set is `WRel` X = `Rel` (`Word` X) 0ℓ. Given Γ : `WRel` X, the library's two-level convention is that `_===_` always means the raw axioms and $\_\approx\_$ the monoid congruence they generate, an inductive family closed under reflexivity, symmetry, transitivity, congruence of $\bullet$, associativity, the unit laws, and `axiom`. Thus $\approx$ *is* the word problem of $\langle X \mid \Gamma \rangle$, and a derivation is a first-class object one can induct on. Because words have no primitive inverses every presentation is at bottom a monoid presentation; a `Grouplike` witness (every generator has a left inverse up to $\approx$) upgrades the presented monoid to the presented group $\bullet$-ε-`group`. The extension of a generator map $f : X \longrightarrow$ `Word` Y to words is written postfix, (f^{w}), and a coset action $h : C \longrightarrow Y \longrightarrow$ `Word` $X \times C$ extends to words as (h^{t}), threading the coset from left to right; these two combinators, with the conjugation extensions $(conj^{\mathrm{h}}{}')$ and $(conj^{\mathrm{n}}{}')$, are the whole syntactic vocabulary of the framework.

A presentation of a concrete group is a record:

```
record _IsPresentationOf_ (_===_ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field
    gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.•-ε-group) (Group.rawGroup G)
  field
    ⟦_⟧ : Word X → Group.Carrier G
    iso : IsGroupIsomorphism ⟦_⟧
```

A weaker record, `_IsSubPresentationOf_`, asks only for a monomorphism; it packages soundness (the homomorphism's congruence) and completeness (its injectivity), both stated with the standard library's `Congruent` and `Injective`, and `isPresentationOf` upgrades it to a presentation given surjectivity. Every presentation theorem in this paper is assembled this way: soundness, completeness by normalisation, and surjectivity are proved separately and the records do the rest.

3.2 Normal forms, and completeness by normalisation

Normal-form witnesses are, on the nose, the standard library's function bundles out of the word setoid: `NormalFormInjective` Γ NF is an `Injection`, `NormalForm` Γ NF a `RightInverse` — the map `nf` together with a section `inv-nf` : $NF \longrightarrow$ `Word` X and a proof `inv-nf`$\circ$`nf` $\approx$ id — and `BijectiveNormalForm` a `Bijection`. The section is data: it is the algorithm that realises each normal form as a canonical word, and it is what completeness consumes. Fix a semantics $[\![\_]\!] : \texttt{Word}\ X \to \mathit{Sem}$ into a setoid. A normal form is *unique for* $[\![\_]\!]$ when representatives with equal denotations are equal normal forms, and *soundness plus a unique normal form yields completeness*:

```
record UniqueNormalForm (inv-nf : |NF| → Word X) : Set (c ⊔ d) where
  field
    unique : ∀ {u v : |NF|} → ⟦ inv-nf u ⟧ ≈₂ ⟦ inv-nf v ⟧ → u ≈ₙ v

by-normalization : {normalForm : NormalForm} → let open NormalForm normalForm in
  UniqueNormalForm inv-nf → Congruent _≈_ _≈₂_ ⟦_⟧ → Injective _≈_ _≈₂_ ⟦_⟧
```

The proof is five lines: given $[\![w]\!] \approx_2 [\![v]\!]$, soundness transports along $w \approx \texttt{inv-nf}\,(\texttt{nf}\,w)$, uniqueness identifies the normal forms, and injectivity of `nf` closes the loop. Its value is that every case study only ever proves `unique` — a statement about normal forms, i.e. about concrete first-order data — never about arbitrary words. This is precisely the division of labour of the qupit paper's Proposition 3.8 (uniqueness of the symplectic normal form) and Proposition 4.3 (every circuit rewrites to normal form), except that the rewriting half is packaged as a *function* with a verified section, and the argument "two circuits with the same symplectic matrix rewrite to the same normal form, hence to each other" is discharged once and for all. A converse, `by-completeness`, recovers `UniqueNormalForm` from completeness plus an exact section $\texttt{nf} \circ \texttt{inv-nf} \equiv \mathrm{id}$; it is the route by which uniqueness witnesses are lifted through products.

3.3 Circuits and the structural rules

Module `Circuit.Base` is parameterised by a gate family $\texttt{Gate} : \mathbb{N} \to \mathrm{Set}$:

```
data Gen : ℕ → Set where
  gate₀ : Gate 0 → Gen n
  gate₁ : Gate 1 → Gen (₁₊ n)
  gate₂ : Gate 2 → Gen (₂₊ n)
  _↥    : Gen n → Gen (₁₊ n)

Circuit : ℕ → Set
Circuit n = Word (Gen n)
```

A generator is a gate at the bottom of the wires shifted up by some `_↥`s; a 0-ary gate is a global scalar, available at every width; $c\!\uparrow = \texttt{wmap}\ \_\!\uparrow$ shifts a circuit. The indices are built from successors only, so that the coverage checker solves its unification problems by injectivity of $_{1+}$ — which is why the coset tables below can be defined by direct pattern matching. Given any gate-specific family of relations, `Lift-Relation` adds the structural rules that every circuit theory shares:

```
module Lift-Relation (_SRel,_===_ : (n : ℕ) → CRel n) where
  data _VRel,_===_ : (n : ℕ) → CRel n where
    srel  : n SRel, w === v → n VRel, w === v
    cong↑ : n VRel, w === v → (₁₊ n) VRel, w ↑ === v ↑
    comm₀ : (h : Gate 0) (g : Gen n) → n VRel,
      [ g ]ʷ • [ gate₀ h ]ʷ === [ gate₀ h ]ʷ • [ g ]ʷ
    comm₁ : (h : Gate 1) (g : Gen n) → (₁₊ n) VRel,
      [ g ↥ ]ʷ • [ gate₁ h ]ʷ === [ gate₁ h ]ʷ • [ g ↥ ]ʷ
    comm₂ : (h : Gate 2) (g : Gen n) → (₂₊ n) VRel,
      [ g ↥ ↥ ]ʷ • [ gate₂ h ]ʷ === [ gate₂ h ]ʷ • [ g ↥ ↥ ]ʷ
```

```
ω↑=ω : (ω : Gate 0) → (₁₊ n) VRel, [ gate₀ ω ]ʷ ↑ === [ gate₀ ω ]ʷ
```

Shift-congruence, far-commutation of disjoint gates, and the identification of a scalar with its shifts: these are the "monoidal" rules that the qupit paper leaves implicit in the phrase "congruence compatible with composition and tensor product", and every rule set in this paper is such a lift. The lemma `lemma-cong↑` lifts the entire congruence one wire up, so equational developments move freely along the wire chain $\mathsf{Cir}\,k \leq \mathsf{Cir}\,(k{+}1)$.

3.4 Coset tables and towers

The library's normal forms all follow one recipe from computational group theory [Holt et al. 2005; Sims 1994]. Given a chain $1 = G_0 \leq G_1 \leq \cdots \leq G_n = G$ with a transversal T_k of each step, every element factors uniquely as $t_1 t_2 \cdots t_n$ — a mixed-radix numeral, one digit per level. For circuits the chain is handed to us by the wire structure: $\mathsf{Cir}\,k$ embeds into $\mathsf{Cir}\,(k{+}1)$ as the circuits that do not touch the bottom wire, and a transversal is a family of *coset representatives* indexed by k. One level is a Reidemeister–Schreier situation, packaged as the module `SingleLevel` with parameters

$$\Gamma : \texttt{WRel}\ X, \quad \Delta : \texttt{WRel}\ Y, \quad C : \mathsf{Set}, \quad I : C, \quad f : X \longrightarrow \texttt{Word}\ Y,$$
$$h : C \longrightarrow Y \longrightarrow \texttt{Word}\ X \times C, \quad [\_] : C \longrightarrow \texttt{Word}\ Y$$

— a subgroup presentation, a group presentation, the cosets, the identity coset, the generator embedding, the *coset table* h that pushes one letter of G past a coset, emitting a word of the subgroup and the successor coset, and the Schreier section. Five hypotheses make the data a genuine coset presentation:

```
(h=⁻¹f-gen : ∀ (x : X) → ([ x ]ʷ , I) ~ ((h ᵗ) I (f x)))
(h-wd-ax : ∀ (c : C){u t : Word Y} → u ===₂ t → ((h ᵗ) c u) ~ ((h ᵗ) c t))
(f-wd-ax : ∀ {w v} → w ===₁ v → (f ʷ) w ≈₂ (f ʷ) v)
([I]≈ε : [ I ] ≈₂ ε)
(h=ract :  ∀ c b → let (b' , c') = h c b in
  [ c ] • [ b ]ʷ ≈₂ (f ʷ) b' • [ c' ])
```

From these, `Transfer` derives the normal-form transport

$$\texttt{nfp'} : \texttt{NormalForm}\ \Gamma\ \mathit{NF} \longrightarrow \texttt{NormalForm}\ \Delta\ (\mathit{NF} \times C) :$$

the new `nf` runs the table from the identity coset and normalises the accumulated subgroup word, the new section is $(f^{\,\mathrm{w}})(\texttt{inv-nf}\ u) \bullet [c]$, and the hypotheses prove they are inverse. `CosetTower` iterates this up an $\mathbb{N}$-indexed family of presentations, one `Extension` record per level, with carrier

$$\texttt{tower-carrier}\ \mathit{NF}_0\ (k{+}1) = \texttt{tower-carrier}\ \mathit{NF}_0\ k \times C_k.$$

The hypotheses are *quantifier-small* — they range over cosets, generators and axioms, all finite inductive types — and in the symmetric group they close by evaluation. In the symplectic group they do not, and §5.5 explains what replaces evaluation there.

3.5 Presentation construction theorems

Relation sets over a disjoint union of alphabets are built from one primitive, a three-way join $\Gamma_1 \diamond \Gamma_2 \diamond \Gamma_3$ whose *mixed* component $\Gamma_3 : \texttt{WRel}\ (A \uplus B)$ is the design parameter. Two mixed components matter here. For the semidirect product the mixed relation says that moving a letter h of the acting group past a letter n of the normal subgroup conjugates n, by a *word*:

```
data ConjRelʷ {N H} (conj : H → N → Word N) : WRel (N ⊎ H) where
  comm : (n : N) (h : H) →
         ConjRelʷ conj ([ [ h ]ʷ ]ᵣ • [ [ n ]ʷ ]ₗ)
```

```
                    ([ conj h n ]₁ • [ [ h ]ʷ ]ᵣ)
```

The semidirect presentation theorem is parameterised by the two relation sets and the action, by two well-definedness conditions on the action — it respects the acting group's axioms in its first argument and the normal subgroup's axioms in its second — and by presentations of the two factors:

```
module Presentation.Construct.Properties.SemiDirectProduct
  {N H : Set} (Γ : WRel N) (Δ : WRel H) (conj : H → N → Word N) where
module _
  (conj-hyph : ∀ {c d} n → c ===₂ d → (conj ʰ') c n ≈₁ (conj ʰ') d n)
  (conj-hypn : ∀ c {w v} → w ===₁ v → (conj ⁿ') c w ≈₁ (conj ⁿ') c v)
  where
  module Presentation (G1 G2 : Group 0ℓ 0ℓ)
    (p1 : Γ IsPresentationOf G1) (p2 : Δ IsPresentationOf G2) where
    dpres : (Γ ⋄ Δ ⋄ ConjRelʷ conj) IsPresentationOf G1⋊G2
```

Here `G1⋊G2` is the semidirect product of the two *semantic* groups, built by the library's `ForStdlib` construction from a bundled `Action` record whose action is the syntactic `conj` transported through the two presentation isomorphisms. Its normal form is the product of the factors' normal forms; the proof is a Reidemeister–Schreier level whose cosets are words over the acting alphabet up to $\approx$, and its uniqueness is lifted from the factors by `by-completeness`. The second mixed component, `RelTwist`, twists each relator $u = v$ of the quotient by a correction word in the normal subgroup, $[u]_r = [corr\ r]_l \bullet [v]_r$, and

```
extension-presentation S R conj corr = S ⋄ EmptyRel ⋄ (ConjRelʷ conj ∪ RelTwist R corr)
```

is the recipe of Proposition 2.55 of the Handbook of Computational Group Theory [Holt et al. 2005], which the qupit paper invokes for its Section 4.2: the relations of the normal subgroup, conjugation rules, and each relator of the quotient corrected by the element of the normal subgroup by which its lift fails to hold. The library proves it as a presentation theorem for any group extension (given a short exact sequence realising the data) and, in the form used in §6.4, for central extensions built from a cocycle. The semidirect product is the special case with trivial corrections.

3.6 Modular arithmetic and primitive roots

Finally, $\mathbb{Z}_p$ itself. `ForStdlib.Data.Fin.Mod` models $\mathbb{Z}_p$ as Fin p with operations on representatives, $\mathbb{Z}_p^*$ as the nonzero residues, inversion by Bézout, and a ring solver; the module `PrimeModulus'` adds Fermat's little theorem, proved by the classical argument that multiplication by a unit permutes the nonzero residues, and `Primitive-Root-Modp'` takes a primitive root g and the witness `g-gen` as parameters and derives from them the discrete logarithm and the exponent-reduction law $g^k = g^{k \bmod (p-1)}$. These are the arithmetic facts behind the one-qupit rules: the multiplier law $M_x M_y = M_{xy}$ is, on exponents, addition modulo $p - 1$, and Fermat is what makes $M_g^{p-1} = \varepsilon$.

4 Two Routes to Completeness

This section summarises the qupit paper's proof, explains why we did not formalise it as written, and describes the route we took instead. The two routes share their mathematics — the same normal boxes, the same symplectic normal form, essentially the same eighteen symplectic relations — and differ in the *organisation* of the completeness argument, which is what a proof assistant cares about.

4.1 The paper's route

The qupit paper [Bian et al. 2026a] proves Theorem 4.10 — the sixteen rules of its Figure 1 are sound and complete for the unitary interpretation — in three moves.

A normal form. Its Section 3 defines a normal form for an n-qupit Clifford circuit as a *symplectic normal form* followed by a *Pauli normal form*: the former, built from the normal boxes A, B, D, E in alternating Z-normal and X-normal layers, represents the stabiliser tableau; the latter, $\bigotimes_j Z^{b_j} X^{a_j}$, is the Pauli correction. Proposition 3.8 shows that symplectic normal forms are in bijection with $\mathrm{Sp}(2n, \mathbb{Z}_p)$, and Proposition 3.13 that every Clifford operator has a unique normal form up to global phase. The proofs (its Appendix E) are "analogous to those for the qutrit case".

Symplectic completeness. Its Section 4.1 rewrites any circuit to symplectic normal form: append the normal form of the identity and push the gates in one by one. Pushing a gate through a normal box yields an updated box and residual *dirty* gates, which are pushed further in (its Figure 7). Collecting every case gives 66 parametrised *box relations* (its Appendix F). The argument that this terminates, and terminates in a clean normal form, is Lemma 4.2, proved by inspecting which dirty gates each box can absorb (its Definition 4.1 and Figure 8), "entirely analogous to" the qutrit paper's Lemma 4.2. Theorem 4.4 then reduces the 66 box relations to 18 symplectic relations (its Figure 9), and here the paper does use Agda: "All proofs have been formalised." Throughout Section 4.1 relations are applied *up to Paulis* — the symplectic interpretation $[\![\_]\!]_s$ identifies a circuit with the same circuit followed by any Pauli — so the box relations, the dirty normal forms and the eighteen rules are all statements about $C_n/\mathcal{P}_n$.

Boosting. Its Section 4.2 lifts symplectic completeness to projective and unitary completeness. Lemma 4.5 says that two circuits with the same symplectic matrix differ by a unique Pauli; Definition 4.7 and Proposition 4.8 say that Figure 1 is *Pauli complete* — it derives the commutation and cancellation identities for Paulis and the six rules for pushing a Pauli through a generator; and Proposition 4.9 is the general boost: given a symplectically complete rule set R_s and a Pauli complete rule set R_p, correct each rule $C_l = C_r$ of R_s to $C_l = C_r\,; P$ for the unique Pauli P that makes it projectively sound, and then $R'_s \cup R_p$ is projectively complete. The proof takes a sequence of symplectic rewrites from C_1 to C_2, replays it with the corrected rules, and after each step pushes the newly created Pauli to the end of the circuit with R_p, where it turns into "a potentially different Pauli"; at the end C_1 has been rewritten to $C_2\,; P$ with $[\![P]\!]_u = \mathrm{id}$, so R_p removes P. The same argument, run once more with $\langle\omega\rangle$ in place of $\mathcal{P}_n$, gives unitary completeness.

4.2 Why we did not formalise that

Each move is correct, and each was stated at exactly the level of precision a careful reader needs. A proof assistant needs more, and the gap is widest at the places where the paper says "analogous", "by inspection" and "potentially different".

First, the rewriting procedure is not a function. "Push the gates in one by one" and "push the dirty gates further in" describe a non-deterministic rewrite relation whose termination is argued by an invariant on the placement of dirty gates (its Definition 4.1), verified by reading the 66 box relations. To formalise Proposition 4.3 one would define the normalisation as a structurally recursive function on circuits, prove it sound with respect to the relations, and prove that its result is a clean normal form — and at that point one has written a coset table, which is what we do (§5.5), except that the coset table also has to say what happens to the Pauli.

Second, and more seriously, the "projective" use of relations in Section 4.1 means that the objects the rewriting manipulates are not circuits but circuits modulo Paulis, without any syntactic representation of the quotient. A box relation $g\,M = M'\ dir$ holds *up to a Pauli* that the relation

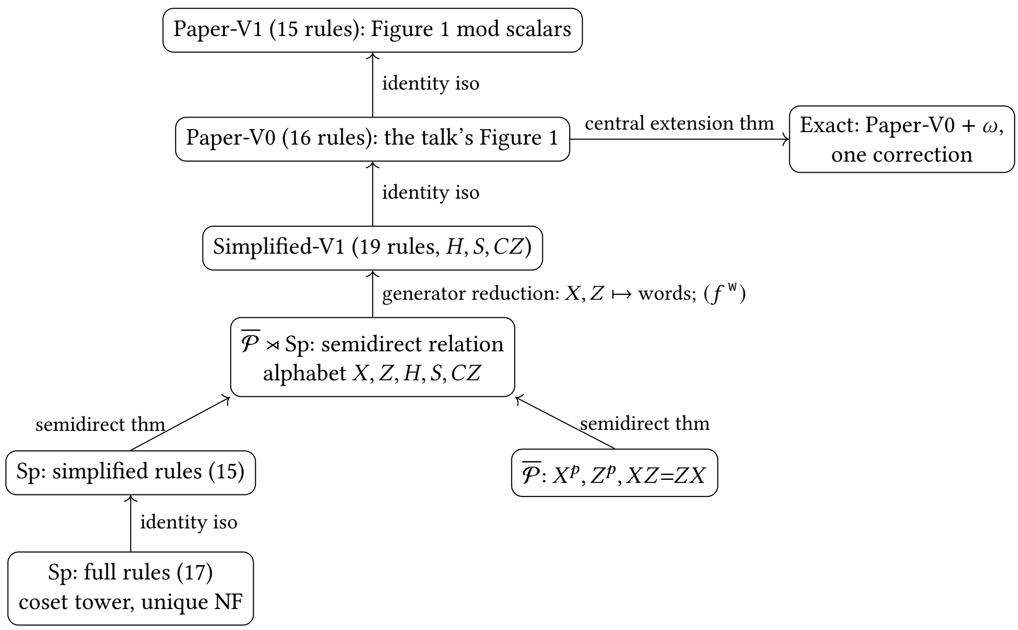


Fig. 3. The chain of presentations. Every arrow is a verified presentation theorem or a verified isomorphism of presentations; the semantic group is $\mathrm{Sp}(2n, \mathbb{Z}_p)$ at the bottom, $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$ from the semidirect level up, and a central extension of it by $\langle\omega\rangle$ at the top right.

does not name. Formalising this faithfully means either carrying the Pauli explicitly through every box relation — turning each of the 66 relations into a relation with a computed Pauli correction, and re-doing the reductions $66 \longrightarrow 18 \longrightarrow 16$ with corrections — or working in a syntax where Paulis are genuinely trivial. The paper does the former informally, in Proposition 4.9, by observing that the corrections *can* be computed and pushed to the end. We found that the honest formal counterpart of that observation is a normalisation procedure on the semidirect product, whose well-definedness (that congruent inputs give equal cosets and congruent residuals) is exactly what the "potentially different Pauli" glosses over: the Pauli produced depends on the rewrite path, and one must prove that all paths agree. We know this works; we did not see how to implement it and verify its properties at reasonable cost.

Third, the uniqueness proof, Proposition 3.8, is by reference to the qutrit paper. Its content is that the action of a normal form on the Pauli generators determines the normal form; formalised, this is a statement about the coset representatives at each level of a tower, and it is where the doubly inductive structure of our normal form pays off (§5.6).

4.3 Our route

We work in a syntax where Paulis are genuinely trivial — the symplectic circuits of §5 have *no* Pauli generators, only H, S, CZ, and their semantics is $\mathrm{Sp}(2n, \mathbb{Z}_p)$ itself — and reintroduce Paulis and scalars afterwards by construction. Figure 3 shows the chain. Reading upward:

(1) The symplectic group is presented by a coset tower whose cosets are the normal boxes of the paper, arranged as a doubly inductive type; the coset table is the paper's "push a gate through a box", now a structurally recursive function with a verified section; the well-definedness of the table is proved *semantically*, by uniqueness of the normal form one

level down; and the box relations the table uses are derived from seventeen full symplectic rules, which are in turn derived from fifteen simplified ones (§5).

(2) The projective Pauli group is presented by three rules and the semidirect product by the generic theorem of §3.5; the only new obligation is that the conjugation action respects both factors' rules, and that is discharged by soundness and completeness of the two factor presentations rather than by hand (§6.2).

(3) "Plumbing" reduces the semidirect alphabet and relations to the paper's: X and Z become words in H and S, the Pauli rules and the conjugation rules become theorems, and two further identity isomorphisms carry the result to the talk's sixteen rules and to the paper's fifteen (§6.3). This is where the two routes meet: *the mod-scalar Figure 1 and the semidirect presentation present the same group.*

(4) The scalars are restored by the extension theorem, with the one correction computed and verified against a cocycle (§6.4).

Does this just move the difficulty around? No. The symplectic factor is where the real rewriting happens, and keeping the Pauli part out of it pays off everywhere: every box relation is an honest equation, every coset-table entry is a function, and the Pauli bookkeeping that the paper's Proposition 4.9 performs by hand is performed once, by the semidirect-product theorem, in a form that was proved for arbitrary groups before this project began. Prior work on the qubit and qutrit Clifford groups [Li et al. 2025; Selinger 2015] worked with the whole Clifford group at once, and their derivations drag a "messy" Pauli correction through every step; the divide-and-conquer route is, we think, the right way to organise such proofs even on paper.

5 The Symplectic Presentation

5.1 Divide and conquer

Factor the group; present the factors; compose the presentations. Two short exact sequences organise the qupit Clifford group:

$$
\begin{aligned}
&1 \to \overline{\mathcal{P}}_n \to \overline{C}_n \to \mathrm{Sp}(2n, \mathbb{Z}_p) \to 1 && \text{semidirect: it splits;}\\
&1 \to \langle \omega \rangle \to C_n \to \overline{C}_n \to 1 && \text{central extension.}
\end{aligned}
$$

This section presents the symplectic factor, which is where the real rewriting happens; §6 presents the Pauli factor and the scalars and composes. The plan for $\mathrm{Sp}(2n, \mathbb{Z}_p)$ has seven steps, which the subsections follow: the normal form; rewriting to it, warming up with S_n; the passage from symmetric to symplectic; uniqueness; relation reduction; the presentation theorem; and a comparison with the qubit case.

5.2 The semantics: symplectic transformations of Pauli vectors

Rather than $2n \times 2n$ matrices, the formalisation models $\mathrm{Sp}(2n, \mathbb{Z}_p)$ as the group of $\mathbb{Z}_p$-linear, symplectic-form-preserving bijections of the phase space `Pauli` n = `Vec` $(\mathbb{Z}_p \times \mathbb{Z}_p)$ n — one pair (a, b), the exponents of $Z^a X^b$ (up to phase), per wire. A symplectic transformation carries its own inverse and its own proofs:

```
record Symplectic (n : ℕ) : Set where
  field
    ap        : Pauli n → Pauli n
    ap⁻¹      : Pauli n → Pauli n
    invˡ      : ∀ p → ap⁻¹ (ap p) ≡ p
    invʳ      : ∀ p → ap (ap⁻¹ p) ≡ p
    linear-+  : ∀ p q → ap (p +ₚ q) ≡ ap p +ₚ ap q
```

```
    linear-*  : ∀ k p → ap (k *ₚ p) ≡ k *ₚ ap p
    preserves : ∀ p q → sform (ap p) (ap q) ≡ sform p q

_≈ˢ_ : ∀ {n} → Symplectic n → Symplectic n → Set
S ≈ˢ T = ap S ≗ ap T
```

Equality is extensional equality of the action alone, so the proof fields never obstruct. Composition is function composition, and therefore associativity and the unit laws hold *on the nose*, by `refl` — a small choice with large consequences downstream, since every monoid case of every soundness proof closes definitionally. `Sp-group` n is the resulting `Group`. The generators act as the qupit paper's conjugation tables (its Lemmas 2.22–2.23) read on exponent vectors:

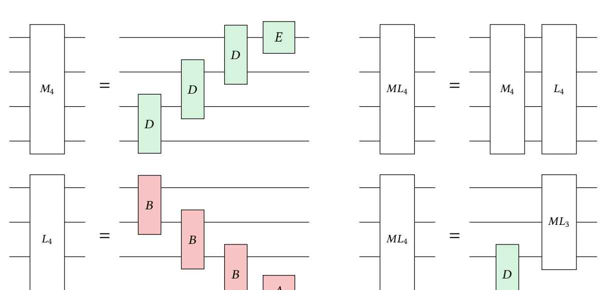
```
actg : ∀ {n} → Gen n → Pauli n → Pauli n
actg (gate₁ H-gate)  ((a , b) ∷ ps)            = (- b , a) ∷ ps
actg (gate₁ S-gate)  ((a , b) ∷ ps)            = (a , b + a) ∷ ps
actg (gate₂ CZ-gate) ((a , b) ∷ (a' , b') ∷ ps) = (a , b + a') ∷ (a' , b' + a) ∷ ps
actg (g ↥)           (x ∷ ps)                  = x ∷ actg g ps

⟦_⟧ : ∀ {n} → Circuit n → Symplectic n
⟦ [ g ]ʷ ⟧ = ⟦ g ⟧ᵍ
⟦ ε ⟧       = εˢ
⟦ w • v ⟧  = ⟦ w ⟧ ∘ˢ ⟦ v ⟧
```

Words are read left to right, so $w \bullet v$ applies w first and its denotation composes on the right. The seven proof obligations per generator — inverses, linearity, preservation of the form — are identities in $\mathbb{Z}_p$, discharged with the ring solver. Soundness of every axiom of §5.7 in this semantics is one clause per axiom in `SoundnessDirect` (1 100 lines).

5.3 The normal form: boxes, rows, and the staircase

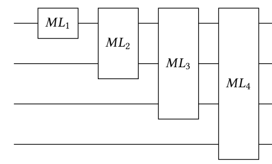
The qupit paper's symplectic normal form is a Z-normal layer followed by an X-normal layer on n wires, then a normal form on $n-1$ wires. We refine this into a doubly inductive datatype. The atomic boxes were shown in §2; rows and the staircase are:

```
M L ML : ℕ → Set

M 0 = ⊤
M (1+ n) = Vec D n × E

L 0 = ⊤
L (1+ n) = Vec B n × A

ML 0 = ⊤
ML 1 = M 1 × L 1
ML m@(2+ n) =
  M m × L m ⊎ D × ML (1+ n)
```



An M-row is a column of D boxes ending in an E box (the paper's X-normal layer); an L-row is a column of B boxes ending in an A box (its Z-normal layer, with the A box at the bottom). An ML box at width $m \geq 2$ is *either* a full pair of rows *or* a bottom D box followed by an ML box one wire up: a spine of D boxes capped by a pair of rows. This is the first induction, on the width of a single coset representative. The second induction is the staircase:

```
NF : ℕ → Set
NF 0 = ⊤
NF (1+ n) = NF n × ML (1+ n)

[_] : ∀ {n} → NF n → Word (Gen n)
[_] {0}      tt         = ε
[_] {1+ n}  (nf , ml)  = [ nf ] ↑ • [ ml ]ᵐˡ
```



Reading normal forms back as circuits is structural recursion, box by box. The box-to-word maps make the pictures of §2 precise; for instance

```
[_]ᵈ : ∀ {n} → D → Word (Gen (2+ n))
[_]ᵈ {n} (0 , b) = Ex • CZ^ (- b)
[_]ᵈ {n} (a@(1+ _) , b) = Ex • CZ^ (- a) • H • S^ -b/a
```

where $-b/a = -b \cdot a^{-1}$, and an A box $A_{a,b}$ with $a \neq 0$ is XM $a \bullet H \bullet S^{-b/a}$, a multiplier, a Hadamard, and an S-power. Why the spine? The paper's Z-normal circuit $W_Z^{(n)}$ has its A box on some wire i, with B boxes above it and nothing below; the position i is an index that the paper's normal-form circuits carry implicitly. The ML type makes it explicit as the length of the D-spine, and, crucially, makes the coset type at width m a *sum* of the coset type at width $m-1$ and a flat case — exactly the shape the coset table and the uniqueness proof recurse on. The identity coset is the flat one with all exponents zero and $A_{0,1}$ at the bottom, and the section sends it to ε up to $\approx$, because $Ex^2 \approx \varepsilon$ collapses each $D_{0,0}$ and $B_{0,0}$.

5.4 Warm-up: coset tables for the symmetric group

Swap circuits on n wires present S_n, and the library's treatment of them [Bian and Feng 2026a] is the template the symplectic development follows. Circuits have a built-in tower of subgroups $\mathrm{Cir}\,0 < \mathrm{Cir}\,1 < \cdots < \mathrm{Cir}\,n$, where $\mathrm{Cir}\,k$ consists of the circuits that touch only the top k wires; for swap circuits this is $S_0 < S_1 < \cdots < S_n$. Representatives of S_3/S_2 (and of S_2/S_1, S_1/S_0) are the descending staircases:

$c_0^{(2)}$ $c_1^{(2)}$ $c_2^{(2)}$ $c_0^{(1)}$ $c_1^{(1)}$ $c_0^{(0)}$

For $f \in S_3$ there is a unique $c \in C_2$ with $c \circ f^{-1}$ fixing the bottom wire — the c that sends $f^{-1}(0)$ to 0 — and recursing on $c \circ f^{-1} \in S_2$ writes every f uniquely as $c^{(0)}c^{(1)}c^{(2)}$: the mixed-radix encoding $|S_3| = 3 \cdot 2 \cdot 1$ of §3.4 (Figure 4, right).

Rewriting to normal form is the coset table. Pushing a generator b through a coset representative $c_3 \in C_3$ yields a new coset c_3' and a residual "dirty" circuit b' on the wires above, which is then pushed through C_2, and so on:

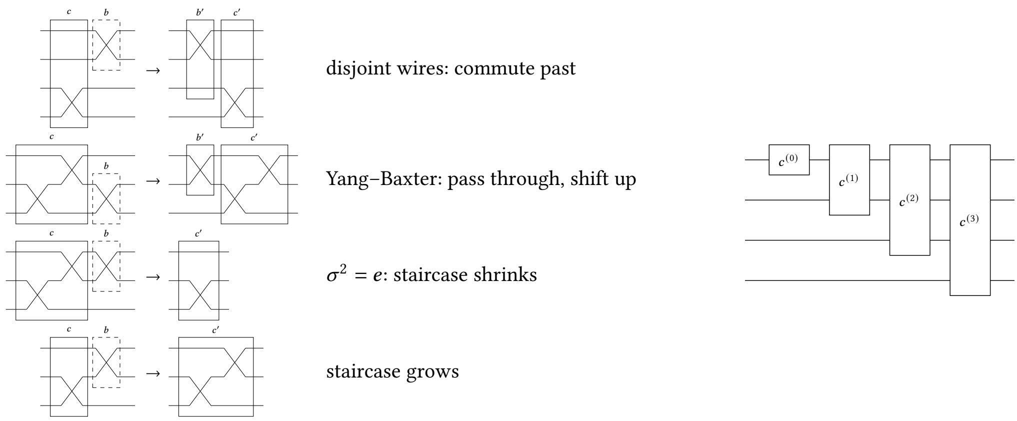
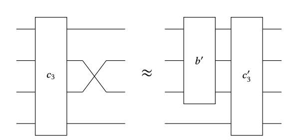


Fig. 4. Left: pushing a swap through a staircase, the four cases of the coset table for S_n. Right: the staircase of staircases, the normal form of a permutation.

```
ract : C (₁₊ n) → Gen (₂₊ n) → Circuit (₁₊ n) × C (₁₊ n)
ract {n}    ε          σ-gen  = ε , σ• ε
ract {n}    (σ• ε)     σ-gen  = ε , ε
ract {₁₊ n} (σ• σ• c) σ-gen  = (σ {n = n}) , σ• σ• c
ract {n}    ε          (g ↥)  = [ g ]ʷ , ε

ract-sound : ∀ {n} c b →
  let (b' , c') = ract {n} c b
  in  [ c ]ᶜ • [ b ]ʷ ≈ b' ↑ • [ c' ]ᶜ
```



The four cases are the four pictures of Figure 4: disjoint wires commute; the Yang–Baxter rule passes the swap through and shifts it up; $\sigma^2 = e$ shrinks the staircase; the staircase grows. Every case has the shape $c \cdot b \approx b' \cdot c'$, the five coset-table hypotheses close by evaluation, and collecting the cases gives a rewrite system that normalises every circuit. Simplifying the collected relations leaves the two Coxeter rules, $\sigma^2 = \varepsilon$ and Yang–Baxter, plus the structural ones, and the library proves that they present the permutation group of Fin n.

5.5 From symmetric to symplectic: change the action

The same framework serves the symplectic group under a richer action. There are three ways of looking at the wires. S_n acts on $\{0, \dots, n-1\}$: a wire is a position, and $c \circ f^{-1}$ fixes 0. S_n acts on Vec Bit n: a wire is a bit, and $c \circ f^{-1}$ fixes the $\mathbb{Z}_2$-linear subspace of the bottom wire. Adding H, S, CZ, the group acts on Vec Pauli_1 n, a $2n$-dimensional symplectic $\mathbb{Z}_p$-vector space in which a wire is a two-dimensional symplectic subspace; enlarging the coset layer from staircases to ML boxes, for each $f \in \mathrm{Sp}(2n, \mathbb{Z}_p)$ there is a c such that $c \circ f^{-1}$ fixes the symplectic subspace of the bottom wire. The cosets at width $1{+}n$ are the ML boxes, their section is $[\_]^{\mathsf{ml}}$, and the coset table is the paper's "push a gate through a normal box", now a function:

```
C = ML

ract : ∀ {n} → C (₁₊ n) → Gen (₁₊ n) → Circuit n × C (₁₊ n)

ract-sound : ∀ {n} c g →
```

```
  let (b' , c') = ract {n} c g
  in [ c ]ᶜ • [ g ]ʷ ≈ b' ↑ • [ c' ]ᶜ
```

`ract` dispatches on the shape of the coset — a flat box or a D-spine — and on the generator — S or H on the bottom wire, CZ on the bottom two, or a lifted gate — and delegates each case to a module of the `Pushing` directory (26 files, 5 600 lines): S and H through a flat box, CZ through a flat box, CZ meeting a D box and a flat box, a lifted gate through the spine. The residual b' lives one wire up, as the type says. The soundness proof `ract-sound` is a chain of ≈-steps per clause, each step a box relation or a structural rule. For example, an S on the bottom wire meeting a spine $D_{a,b}$ followed by lm: S commutes past the lifted lm, then the box relation $D \leftarrow S$ rewrites $D_{a,b} \bullet S$ to $dir{\uparrow} \bullet D_{a,b-a}$, and re-association finishes.

So far this is the S_n story with bigger cosets. The difference is in the five hypotheses of §3.4. Four close as before; the second, `h-wd-ax` — that the table respects the axioms, i.e. congruent inputs produce equal cosets and congruent residuals — does *not* close by evaluation, because the axioms of $\mathrm{Sp}(2n, \mathbb{Z}_p)$ (unlike $\sigma^2 = e$) are not decided by a finite computation on the coset data. The formalisation proves it in two halves. The *coset half* is proved semantically: apply $[\![\_]\!]$ to `ract-sound` on both sides of an axiom, observe that residuals live one wire up and so cannot move the bottom wire, and conclude from head-injectivity of the coset section (§5.6) that the two cosets are equal. The *residual half* then follows by cancellation — provided the shift ↑ is injective on congruence classes at the lower width, which is a consequence of faithfulness (completeness) there. This makes the tower an induction:

```
record TowerLevel (n : ℕ) : Set where
  field
    nfp'  : NormalForm (n QRel,_===_) (NFᵗ n)
    agree : ∀ u → PB._≈_ (n QRel,_===_) (SNF.NormalForm.inv-nf nfp' u) [ φ u ]

-- The induction.  Level (1+k) needs ↑-injectivity at k, which comes
-- from faithfulness at k, which comes from Level k -- strictly below,
-- so the recursion is well founded.
tower : ∀ n → TowerLevel n
tower 0 = record { nfp' = base0' ; agree = λ _ → PB.refl }
tower (₁₊ k) = record { nfp' = nfp'₁ ; agree = agree₁ }
  where
  prev = tower k
  ↑inj = SemInj.Injectivity!.↑-inj-↑ k
           (FF.faithful-from k (TowerLevel.nfp' prev) φ φ-inj (TowerLevel.agree prev))
  nfp'₁ = T.Extension.nfp' (ext k ↑inj) (TowerLevel.nfp' prev)
```

The loop is: a normal form at width k gives faithfulness at k (`faithful-from'`, by `by-normalization`); faithfulness gives injectivity of ↑ on congruence classes (`SemInj`, using that a lifted word acts as $\mathrm{id} \oplus [\![w]\!]$); that discharges `h-wd-ax` at width k; and the generic `Extension.nfp'` produces the normal form at width $k{+}1$. The base case, width 1, is built by hand. This is the formal counterpart of the paper's Lemma 4.2 and Proposition 4.3: no termination argument is needed beyond structural recursion, and "the dirty gates eventually get removed" is the type of `ract`.

5.6 Uniqueness of the normal form

Theorem 2.2 says that normal forms with equal denotations are equal. The action of a normal form on a Pauli vector peels the staircase one wire at a time — the coset factor acts, the resulting bottom entry is fixed, and the inner normal form acts on the tail — and injectivity is a structural induction on the staircase:

```
act-nf : ∀ {n} → NF n → Pauli n → Pauli n
act-nf {0}     _        = id
act-nf {₁₊ n} (ih , lm) ps = head s ∷ act-nf ih (tail s)
  where s = act [ lm ]ᵐˡ ps

lemma-lm-head-inj : ∀ {n} (lm₁ lm₂ : ML (₁₊ n)) →
  (∀ (ps : Pauli (₁₊ n)) → head (act [ lm₁ ]ᵐˡ ps) ≡ head (act [ lm₂ ]ᵐˡ ps)) → lm₁ ≡ lm₂

lemma-lm-tail-surj : ∀ {n} (lm : ML (₁₊ n)) (qs : Pauli n) →
  ∃ λ (ps : Pauli (₁₊ n)) → tail (act [ lm ]ᵐˡ ps) ≡ qs

lemma-nf-inj : ∀ {n} (nf₁ nf₂ : NF n) → act-nf nf₁ ≗ act-nf nf₂ → nf₁ ≡ nf₂
```

Two obligations. *Tail surjectivity* is free, because a symplectic transformation carries its inverse. *Head injectivity* — the bottom-wire output determines the coset — is the crux, and it is itself an induction on the *ML* type (`LMHeadInj`, 1 300 lines): at width 1 a case analysis on the four shapes of an *A* box using closed forms for the actions of S^k, HS^k and the multipliers; a separation lemma showing a flat box and a spine never agree on heads; and, for two spines, the *D* box is read off by probing with the two basis Paulis Z and X on the bottom wire and the identity above, after which the hypothesis descends one width. With these two lemmas the generic `Circuit.Uniqueness` module of the library assembles `[[[]]]-injective` and the `UniqueNormalForm` witness; the module's header records the one subtlety, that `ap` is a *left* action — `ap` of a composite applies its right factor first — so the coset factor of a normal form is the one next to the input, which is what lets the coset be recovered from every state rather than from states of a restricted shape. This is the paper's Proposition 3.8 with its Lemmas 3.4–3.7, now an induction whose structure is dictated by the datatype.

5.7 Relation reduction: box relations and the two rule sets

Which relations does `ract-sound` use? Rewriting *every* circuit to normal form and collecting every relation the rewrite ever uses gives the paper's box relations — one per gate per box shape, parametrised by the box's labels. The formalisation organises them as nine families of case tables, each a lemma quantified over the box parameters with a computed direction `dir-of` and updated box:

wires	family	pushed
1	$A \leftarrow H, A \leftarrow S, E \leftarrow S$	H or S through an A box; S through an E box
2	$B \leftarrow H{\uparrow}, S{\uparrow}, S;\ D \leftarrow H, S{\uparrow}, S, CZ;\ L \leftarrow CZ$	a gate through a B, a D, or an L-row
3	$BB \leftarrow CZ{\uparrow}, B{\uparrow} \leftarrow CZ, DD \leftarrow CZ$	CZ through two B or two D boxes

For instance, the two-wire table for a *D* box is

```
D⟵H-S↥-S-CZ : ∀ (d : D) (g : Gen 2) (neq : g ≢ H-gen ↥) →
  let (e , dir) = TD.dir-of d g neq ; d' = TD.d'-of d g neq
  in [ d ]ᵈ • [ g ]ʷ ≈ S^ e • dir ↑ • [ d' ]ᵈ

d'-of (a , b) H-gen _       = (b , - a)
d'-of (a , b) S-gen _       = (a , b + - a)
d'-of (a , b) (S-gen ↥) _   = (a , b)
d'-of (a , b) CZ-gen _      = (a , b + - ₁)
```

— the paper's Figure 18, with the residual split into an *S*-power on the bottom wire and a word one wire up, and with the box update a function of the labels. The nine families are the paper's

Figures 15–21; unfolding the case tables gives 42 parametric relations, against the paper's count of 66, the difference being only in how cases are grouped (the formalisation, for instance, proves each of the ↑/↓ pairs of a B box and a D box once and obtains the other by a duality lemma). The width-3 relations are stated at width 3 and widened to arbitrary width by a congruence `cong↓ᵏ`.

The box relations are derived (BR, 7 000 lines, on top of a calculus of Ex-conjugations and two-qupit lemmas of 11 000 lines) from the *full* symplectic rule set, seventeen group-specific axioms whose two- and three-qupit members are Selinger's c10–c15 [Selinger 2015] verbatim and whose one-qupit members are the multiplier calculus:

```
order-S     : (₁₊ n) SRel,  S ^ p === ε
order-H     : (₁₊ n) SRel,  H ^ 4 === ε
order-SH    : (₁₊ n) SRel,  (S • H) ^ 3 === ε
comm-HHS    : (₁₊ n) SRel,  H • H • S === S • H • H
M-mul       : ∀ x y → (₁₊ n) SRel,  M x • M y === M (x *' y)
semi-MS     : ∀ x →   (₁₊ n) SRel,  M x • S === S^ (x ^2) • M x
semi-M↑CZ   : ∀ x →   (₂₊ n) SRel,  M x ↑ • CZ === CZ^ (x ^1) • M x ↑
semi-M↓CZ   : ∀ x →   (₂₊ n) SRel,  M x ↓ • CZ === CZ^ (x ^1) • M x ↓
```

together with `order-CZ`, `comm-CZ-S↓`, `comm-CZ-S↑` and `selinger-c10` to `c15`. Here M_x is spelled $S^xHS^{x^{-1}}HS^xH$, and four of the axioms are *schemes* over units x, y. By inspection one finds a smaller set of primitive, bidirectional relations that derive all of these: the *simplified* symplectic rule set, which replaces the schemes by statements about the single multiplier M_g of the primitive root, strengthens $H^4 = \varepsilon$ to $H^2 = M_{-1}$, and drops `order-SH` and `comm-HHS`:

```
order-S :             (₁₊ n) SRel,  S ^ p === ε
order-H :             (₁₊ n) SRel,  H ^ 2 === M₋₁
M-power :      ∀ k → (₁₊ n) SRel,  Mg^ k === M (g^ k)
semi-MS :             (₁₊ n) SRel,  Mg • S === S^ (g * g) • Mg
semi-M↑CZ :           (₂₊ n) SRel,  Mg ↑ • CZ === CZ^ g • Mg ↑
semi-M↓CZ :           (₂₊ n) SRel,  Mg ↓ • CZ === CZ^ g • Mg ↓
```

plus the same `order-CZ`, `comm-CZ-S↓`, `comm-CZ-S↑` and `selinger-c10` to `c15`: fifteen group-specific rules, or eighteen with the three structural ones, shown as circuits in Figure 5. The choice is not canonical; one picks the set one finds natural. The two rule sets generate the same congruence, and since they share an alphabet the identity on words is an isomorphism of presentations in both directions (`Simplified.Iso`); the substantive direction derives each full axiom from the simplified ones. The multiplier law is typical:

```
M-mul : ∀ n (x y : ℤ* ₚ) → let open PB (Sim._QRel,_===_ (₁₊ n)) in
        M x • M y ≈ M (x *' y)
M-mul n = SimL.Lemmas1.lemma-M-mul n
```

— $-\boxed{M_x}-\boxed{M_y}- \approx -\boxed{M_{xy}}-$ — write $x = g^k$, $y = g^l$, use `M-power` twice, add the exponents, reduce modulo $p - 1$ using $M_g^{p-1} \approx \varepsilon$, which is Fermat's little theorem (§3.6), and use `M-power` once more. The whole reduction from seventeen rules to fifteen is 900 lines; the reduction from the box relations to the seventeen is 18 000. This is the formal counterpart of the paper's Theorem 4.4, whose proof the paper already delegated to Agda.

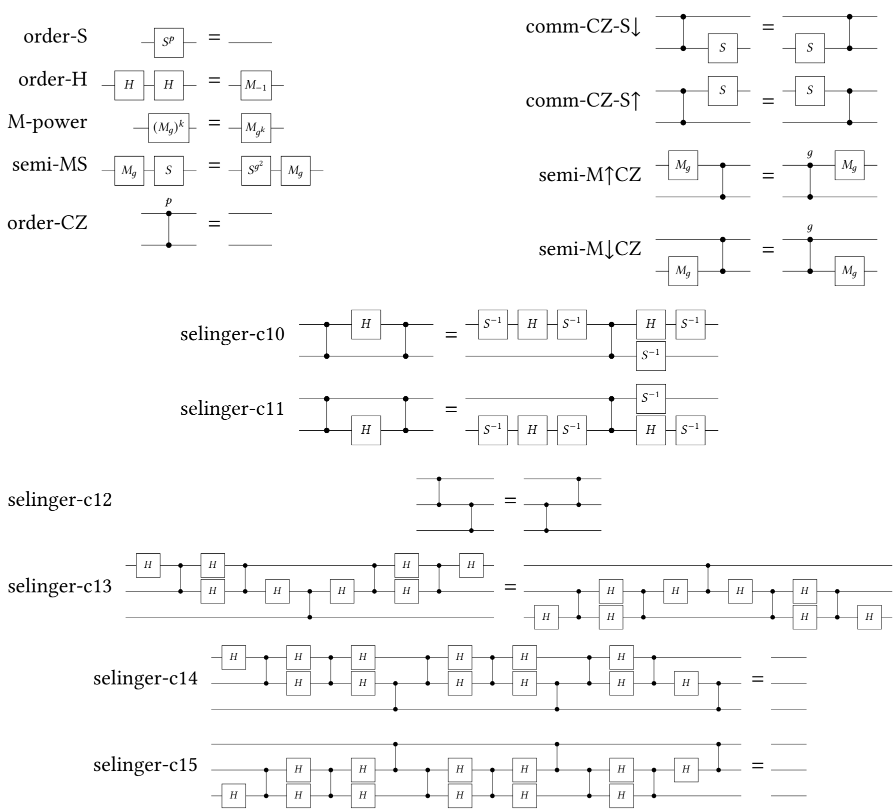


Fig. 5. The simplified symplectic rule set: fifteen group-specific rules (here M_g and M_{-1} abbreviate the *SHS*-words of the text).

5.8 The symplectic presentation theorem

Soundness (one clause per axiom), the normal form of §5.5 and its uniqueness (§5.6) assemble, through `by-normalization`, into a sub-presentation: the interpretation is an injective homomorphism from the words modulo the full rules into `Sp-group`. Surjectivity — every symplectic transformation is the action of some circuit — is proved constructively by "clear the bottom wire, recurse on the tail": the single-level input is

```
Theorem-LM : ∀ {n} (p q : Pauli n) → sform p q ≡ 1 →
  ∃ λ (lm : ML n) → act [ lm ]ml p ≡ pZ0 × act [ lm ]ml q ≡ pX0
```

— for any symplectic pair there is a coset box sending it to the bottom-wire Z and X — which is the paper's Lemma 3.7 with the box computed rather than asserted. Composing the sub-presentation of the full rules with the identity isomorphism of §5.7 gives the theorem for the simplified rules, with the *same* interpretation:

```
Theorem-Sp : ∀ {n} → (n SRel,_===_) IsPresentationOf (Sp-group n)
Theorem-Sp = SimPres.presentation
```

5.9 Comparison with the qubit case

Selinger's Figure 8 [Selinger 2015] presents the qubit Clifford group with the scalars and Paulis included. Read modulo Paulis and scalars, its three-qubit relations are *identical* to `selinger-c12` to `c15` — the reason those names were kept — and its two-qubit relations are *similar*, with S^{-1} for S and CZ^{p-1} for CZ. The one-qupit relations are *more complicated*: the odd-prime multiplier calculus of M_g, with its dependence on a primitive root and on Fermat's theorem, has no qubit counterpart, where the only multiplier is $M_{-1} = H^2$ up to phase. That the difficulty of the general odd prime concentrates on one wire is what makes the divide-and-conquer route affordable: the two- and three-wire rewriting is Selinger's, and the one-wire arithmetic is arithmetic.

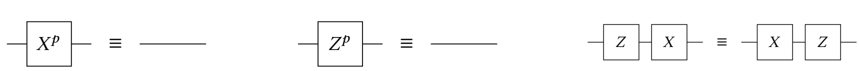
6 Composing: Paulis, the Semidirect Product, and Scalars

From the symplectic presentation to the Clifford presentation there are four steps: the Pauli factor; the semidirect product, gluing the Pauli and symplectic presentations along the conjugation action; plumbing, which simplifies the glued presentation to the paper's rules over H, S, CZ alone; and the central extension by $\langle\omega\rangle$, which restores the scalars.

6.1 The Pauli factor

The projective Pauli group $\overline{\mathcal{P}}_n \cong (\mathbb{Z}_p \times \mathbb{Z}_p)^n$ has an easy presentation over generators X, Z on each wire:

```
data _PRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-X   : (1+ n) PRel,  X ^ p === ε
  order-Z   : (1+ n) PRel,  Z ^ p === ε
  comm-Z-X  : (1+ n) PRel,  Z • X === X • Z

Theorem-Pauli : ∀ {n} → (XZ._QRel,_===_ n) IsPresentationOf (+p-group n)
Theorem-Pauli = XZ.presentation
```



The semantic group $+_p$-group n is `Vec` $(\mathbb{Z}_p{\times}\mathbb{Z}_p)$ n under componentwise addition — the very phase space on which $\mathrm{Sp}(2n, \mathbb{Z}_p)$ acts in §5.2, which is what will make the semidirect action definable. The normal form is $X^a Z^b$ on each wire, read off the exponent vector; soundness, uniqueness and `by-normalization` give the presentation, and the module also exports the soundness and completeness maps separately, which the next step consumes. (The library also presents the same group compositionally, as an n-fold direct product of $\mathbb{Z}_p \times \mathbb{Z}_p$ with no normal-form work at all; that presentation is over a different alphabet, and the circuit-alphabet one is the one the qupit chain needs.)

6.2 The semidirect product

A semidirect product $N \rtimes H$ is presented by the relations of both factors plus twisted commutation: $h\,n\,h^{-1} = \mathit{conj}\,h\,n$, so that $(n,h)\cdot(n',h') = (n\,(\mathit{conj}\,h\,n'),\ h\,h')$. Here the action of a symplectic generator on a Pauli generator is the paper's Lemmas 2.22–2.23, as words — H swaps X and Z (up to an inverse), S sends X to XZ, CZ couples the two bottom wires, and a shifted gate acts on a shifted Pauli only:

```
conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)
conj Sym.H-gen   X-gen           = Z
conj Sym.H-gen   Z-gen           = X ^ p-1
```

```
conj Sym.S-gen   X-gen            = X • Z
conj Sym.S-gen   Z-gen            = Z
conj Sym.CZ-gen  X-gen            = X • Z ↑
conj Sym.CZ-gen  Z-gen            = Z
conj Sym.CZ-gen  (X-gen ↥)        = X ↑ • Z
conj Sym.CZ-gen  (Z-gen ↥)        = Z ↑
conj Sym.H-gen   (xz ↥)           = [ xz ]ʷ ↑
conj Sym.S-gen   (xz ↥)           = [ xz ]ʷ ↑
conj Sym.CZ-gen  (xz ↥ ↥)         = [ xz ]ʷ ↑ ↑
conj (sym ↥)     X-gen            = X
conj (sym ↥)     Z-gen            = Z
conj (sym ↥)     (xz ↥)           = conj sym xz ↑

Pauli⋊Symplectic : (n : ℕ) → WRel (XZ.Gen n ⊎ Sym.Gen n)
Pauli⋊Symplectic n = (n PRel,_===_) ◇ (n SpRel,_===_) ◇ ConjRelʷ conj
```

(module qualifiers on the Pauli side elided; the relation is `_QRel,_===_` in the source).

The generic theorem of §3.5 asks for two things beyond the factor presentations: that `conj` respects the symplectic rules in its first argument and the Pauli rules in its second. Neither is proved by hand. Both go through the semantics: the action of a symplectic word c on a Pauli word w, computed syntactically by $(\mathit{conj}^{h}{}')$, denotes `ap` $[\![c]\!]$ $(\mathit{sem}\ w)$ in the phase space (`conjw-sem`), so if $c \approx d$ by a symplectic axiom then the two conjugates have equal denotations by symplectic soundness, and are therefore congruent Pauli words by Pauli completeness:

```
respects-Δ : ∀ {n} {c d : Word (Gen n)} (u : Word (XZ.Gen n)) →
             Sim._QRel,_===_ n c d →
             PB._≈_ (XZ._QRel,_===_ n) ((conj ʰ') c u) ((conj ʰ') d u)
respects-Δ {n} {c} {d} u ax = complete n
  (Eq.trans (conjw-sem c u)
    (Eq.trans (denote-eq ax (sem u)) (Eq.sym (conjw-sem d u))))
```

and symmetrically for `respects-Γ`. As the module's header puts it, the long symplectic rules — `M-power`, `semi-M↑CZ`, `selinger-c10` to `c15` — are never conjugated by hand. This is the first place the divide-and-conquer route pays off in a way the paper's route cannot: the paper's "Pauli complete" rules (its Definition 4.7) are precisely these conjugation rules, and there they must be derived syntactically from Figure 1 (its Proposition 4.8). With every premise a theorem, the presentation is unconditional:

```
module Semidirect (n : ℕ) where
  private module P = SDP.Presentation (CA.respects-Δ {n}) (CA.respects-Γ {n})
                       (+ₚ-group n) (Sp-group n)
                       (XZ.presentation {n}) (SimPres.presentation {n})

  Pauli⋊Sp : Group 0ℓ 0ℓ
  Pauli⋊Sp = P.G1⋊G2

  presentation : (SemiDirect._QRel,_===_ n) IsPresentationOf Pauli⋊Sp
  presentation = P.dpres
```

The semantic side. `Pauli⋊Sp` is the library's semidirect product of the two semantic groups, built from an `Action` record (§3.5) whose action is the syntactic `conj` transported through the two presentation isomorphisms: $act\ g\ x = [\![conj\ (inv_2\, g)\ (inv_1\, x)]\!]_1$, where inv_i pick words representing g and x. The *natural* semidirect product, with $\mathrm{Sp}(2n, \mathbb{Z}_p)$ acting on $\mathbb{Z}_p^{2n}$ by `ap` — the `Action` whose `act-·-homo` is `linear-+` and whose identity and composition laws are `refl` — is a different term, and the two are identified by a lemma `act≡ap` proving that the transported action is `ap`, so that the multiplications agree (`·-agrees`). This lemma lives in an `abstract` block: with it transparent, the file's header records, a two-line lemma about Z^j "went from 27 s to out-of-memory at 12 GB", because Agda tried to unfold the transported action through both presentation isomorphisms. The semidirect-product structure also implies that the projection to $\mathrm{Sp}(2n, \mathbb{Z}_p)$ is a homomorphism — the symplectic operators embed into the projective Clifford operators as the words over H, S, CZ — so the symplectic relations lift to Clifford relations unchanged.

6.3 Plumbing: presentation simplification

Two presentations are *equivalent* when the presented groups are isomorphic — when each side's axioms are derivable from the other's. The semidirect presentation has five generators per wire, X, Z, H, S, CZ, and the relations of Table 2; the paper has three, H, S, CZ, and sixteen relations. Getting from one to the other is *generator reduction* — X and Z become the derived words of §2 — and *relation reduction* — the Pauli rules and the fourteen conjugation rules become derivable. For example, $CZ \bullet X\uparrow \approx X\uparrow \bullet Z\downarrow \bullet CZ$ (`conj CZ-gen (X-gen ↥)`) becomes a theorem of the Clifford rules.

Table 2. The three presentations of the projective Clifford group and the two reductions between them.

	`SemiDirect`	`Simplified-V1`	`Paper-V0` / `Paper-V1`
generators	X, Z, H, S, CZ	H, S, CZ	H, S, CZ
relations	Pauli (3) + symplectic (15) + conjugation (14)	19	16 / 15
generator reduction: X, Z become the derived words; $S \mapsto R = S \cdot Z^{1/2}$			
relation reduction: Pauli + conjugation rules become derivable; c10–c15 and the swap calculus interderivable			

From the semidirect relation to Simplified-V1. The first reduction is an isomorphism of presentations on *different* alphabets, given by generator maps in both directions:

```
f : ∀ {n} → SemiDirect.Gen n → Word (Gen n)       -- semidirect → Clifford
f X-gen  = Clifford.X
f Z-gen  = Clifford.Z
f H-gen  = H
f S-gen  = Clifford.R
f CZ-gen = CZ

h : ∀ {n} → Gen n → Word (SemiDirect.Gen n)       -- Clifford → semidirect
h H-gen  = SemiDirect.H
h S-gen  = SemiDirect.Z^ -1/2 • SemiDirect.S
h CZ-gen = SemiDirect.CZ
```

(both maps commute with ↥). The symplectic S is carried to $R = S \cdot Z^{1/2}$ and h compensates, because the target rule set, `Simplified-V1`, spells the multiplier over R, as the talk does: $M_x = R^x H R^{x^{-1}} H R^x H$ exactly, with no Pauli correction. `Simplified-V1` has nineteen group-specific rules:

the simplified symplectic rules of §5.7 with R in place of S in the multiplier rules and in c10, c11, plus $(SH)^3 = \varepsilon$, $H^2SH^2S = SH^2SH^2$, and the two rules for pushing X through CZ. Notably $XZ = ZX$ is *not* among them; it is derived from $(SH)^3 = \varepsilon$ and `M-power`, by the paper's own argument (its Proposition 4.8). The isomorphism theorem needs the four generator-level facts of the library's `StarGroupIsomorphism`: f and h respect the axioms (`f-well-defined`, `h-well-defined`, one clause per axiom, the former containing the derivations of all three Pauli rules and all fourteen conjugation rules), and each is a left inverse of the other on generators. Composing with the semidirect theorem:

```
presentation : ∀ {n} →
  (Clifford-Relations._QRel,_===_ n) IsPresentationOf (Semidirect.Pauli⋊Sp n)
```

This is where the mathematics of the composition step lives (about 6 000 lines): a multiplier calculus, a Pauli-conjugation calculus, the CZ layer, and a small rewriting tactic — a commutation oracle for the generators and a step function of some sixty clauses — that discharges the bookkeeping the paper's hand derivations spend most of their length on.

From Simplified-V1 to the talk's sixteen rules. `Paper-V0` is the rule set of §2: the same alphabet, so the identity on words is the candidate isomorphism, and the theorem is again `StarGroupIsomorphism` with `id`. Ten of the axioms are shared verbatim. The rest is real work in both directions. `Paper-V0` axiomatises the swap directly (`order-Ex`, `semi-Ex-S↑`, `semi-Ex-H↑`, `yang-baxter`, `cz-slide`) and two-qupit interactions through `blake-c12` and `semi-CX↑-CZ↓`; from these its `Lemmas` (4 750 lines) derives Selinger's c10–c15 natively — c12 by cancellation against $Ex \bullet Ex\!\uparrow$, c13 because both sides are CZ_{02}, c14 because `⊤⊥↑` has order three so the relation telescopes, c15 as c14 transported along the transposition of wires 0 and 2 — together with the lower-wire rules, the two Pauli-versus-CZ rules and the swap–CZ commutation. Conversely the seven swap rules are proved in `Simplified-V1` by transport along the chain symplectic → simplified → semidirect → V1 rather than by fresh derivations. The module exports

```
v1⇒pap : ∀ {w v} → let open PB (V1R._QRel,_===_ n) using (_≈_) in
         w ≈ v → let open PB (PapR._QRel,_===_ n) renaming (_≈_ to _≈'_) in w ≈' v
```

which transports any Simplified-V1 theorem into the talk's theory; it is what lets later layers reuse the calculi rather than re-prove them.

From the talk's sixteen rules to the paper's fifteen: connecting the two routes. The paper's Figure 1 spells the multiplier over S, with an explicit Pauli prefix (its Lemma 2.14):

```
SHS' a b = Z^ ((b + - ₁) * 1/2) • X^ ((₁ + - a) * 1/2)
           • S^ b • H • S^ a • H • S^ b • H
M x  = SHS' x⁻¹ x
```

and correspondingly states `semi-MR` as $M_g \bullet S^{g^2} \bullet Z^{(g^2-g)/2} = S \bullet M_g$ (its rule C4), the Z-exponent being what is left after the $Z^{1/2}$ hidden in each R is split off and carried across the multiplier, where it picks up a factor of g. `Paper-V1` is Figure 1 modulo scalars in this spelling: the talk's sixteen rules with `order-H`, `M-power`, `semi-MR` and `semi-M↑CZ` restated over `SHS'`, and *fifteen* rules rather than sixteen, because $(SH)^3 = \varepsilon$ is now a theorem:

```
lemma-M1-unfold : M₁ {n} ≈ (S • H) ^ 3
lemma-order-SH  : (S • H) ^ 3 ≈ ε
lemma-order-SH  = trans (sym lemma-M1-unfold) lemma-M1
```

Under the S-spelling, M_1 *is* the word $(SH)^3$ — at $x = 1$ both prefix exponents are $(1-1)/2 = 0$ and the Pauli prefix vanishes — and `M-power` at $k = 0$ already gives $M_1 \approx \varepsilon$. Under the talk's R-spelling M_1 is $(RH)^3$, a different word, which is why the axiom had to stay there. The isomorphism with

`Paper-V0` is again the identity on words; eleven axioms cross directly, and the four multiplier rules need the bridge between the two spellings, $R^a H R^b H R^a H \approx$ `SHS' a b`, which is proved once, over an abstract relation, in `Shared.PauliBase` — a Pauli calculus resting on the axioms both spellings share, with $XZ \approx ZX$ as a *hypothesis*, the one fact whose proof genuinely differs between them — and transported into `Paper-V0`'s theory by `v1⇒pap`. The result is the theorem the index module states:

```
clifford-presentation :
  ∀ n → (PapV1.Clifford-Relations._QRel,_===_ n) IsPresentationOf (Pauli⋊Sp n)
```

So the paper's Figure 1, read modulo scalars, presents the same group as the semidirect presentation of §6.2: the two routes — the paper's single rule set with its projective use of relations, and our factored construction — provably arrive at the same place.

6.4 Getting the scalars back

The Clifford group is a central extension of the projective Clifford group by the scalars $\langle\omega\rangle$. Following the extension recipe of §3.5, the projective presentation is modified in four moves: add a generator ω; add the relation $\omega^p = \varepsilon$; add relations that ω commutes with every gate; and *correct* each projective relation by the scalar it actually picks up. In Agda the recipe is instantiated rather than spelled out:

```
Scalar-relation : WRel ScalarGen
Scalar-relation = p Cn,_===_                -- ⟨ ω | ωᵖ = ε ⟩, cyclic

conj : Gen n → ScalarGen → Word ScalarGen
conj _ _ = ω                                 -- ω is central

sh-exponent : ℕ
sh-exponent = (p Nat.* p Nat.∸ 1) DM./ 8

srel-corr : ∀ {u v} → (n CB.SRel, u === v) → Word ScalarGen
srel-corr CB.order-SH = ω ^ sh-exponent
srel-corr _           = ε

_Exact,_===_ : (n : ℕ) → WRel (ScalarGen ⊎ Gen n)
_Exact,_===_ n =
  extension-presentation Scalar-relation (n CR.QRel,_===_) conj corr
```

Reading the generators as the paper's matrices, with H normalised by δ_p so that all determinants are one, exactly *one* of the sixteen rules fails to hold on the nose: $(SH)^3 = \varepsilon$ picks up $\omega^{(p^2-1)/8}$ (cf. the phase in the paper's (T1)), which is exact division since $8 \mid p^2 - 1$ for odd p, and whose residue modulo p is $-1/8$ — so the corrected rule reads

$$\boxed{S}\,\boxed{H}\,\boxed{S}\,\boxed{H}\,\boxed{S}\,\boxed{H} \equiv \omega^{-1/8}$$

where $1/8$ is the inverse of 8 in $\mathbb{Z}_p$. Every other rule holds exactly: the three multiplier rules and `order-H` because $R = S \cdot Z^{1/2}$ has no Pauli part, `blake-c12` because every Pauli in it is pure-Z, and pure-Z Paulis have vanishing symplectic form. Note that the quotient here is the talk's sixteen-rule set; correcting the paper's fifteen would give Figure 1 itself.

The semantic side. A central extension of a group Q by an abelian group K is determined by a 2-cocycle $c : Q \times Q \to K$, with multiplication $(k, g) \cdot (k', h) = (k + k' + c(g, h),\ gh)$. The formalisation builds two. The *structural* one is the Weyl cocycle on the natural semidirect product,

```
cocy (P , S) (Q , _) = 1/2 * sform P (ap S Q)

weyl : Cocycle Scalar-abelianGroup (Pauli⋊Sp-group n)
Clifford-group = CE.group Scalar-abelianGroup (Pauli⋊Sp-group n) weyl
```

whose cocycle law is symplectic invariance of `sform`, and which satisfies the Heisenberg law $P \cdot Q = \omega^{\mathrm{sform}(P,Q)}\, Q \cdot P$ on Paulis — what distinguishes it from the direct product $\langle\omega\rangle \times (\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p))$. The *presented* one, `Presented-group`, has as quotient the word group of the sixteen rules and as kernel the word group of $\langle \omega \mid \omega^p \rangle$, and its cocycle is generated from *generator data*: for each gate a and word v, the scalar by which the lifts of a and v fail to compose, together with the proof that this descends to the quotient and assigns each relator the same correction on both sides. The generator data is obtained by pulling the Weyl cocycle back along the projective presentation theorem, so the two cocycles agree by construction. The theorem, again on no hypothesis, is

```
presentation-exact : ∀ (n : ℕ) →
  let Clifford-group = λ n → PE.Presented-group n (exact-generator-data n)
  in (n Exact,_===_) IsPresentationOf (Clifford-group n)
```

Its proof needs no normal forms — since both factors are word groups, $[\![\_]\!]$ is a splitting rather than a normalisation — but it does need, for each of the sixteen rules, that the phase the Weyl cocycle assigns to the left-hand side equals the correction plus the phase of the right-hand side. This reduces to nineteen equations in $\mathbb{Z}_p$ about an explicit phase function Φ on circuits: ten are Pauli arithmetic; the shift case says Φ ignores wire shifts; the three multiplier rules and `order-H` are discharged by a one-wire calculus that reads a circuit as a triple (Pauli, 2×2 symplectic matrix, phase) with an explicit concatenation law, so that Φ is *computed*; `blake-c12` by the pure-Z observation; and `order-SH` by the number-theoretic fact that the residue of $(p^2 - 1)/8$ is $-1/8$. The extension is provably non-split: a trivial cocycle could realise the corrections only if $\omega^{(p^2-1)/8} = \varepsilon$, and it is not. This is the formal counterpart of the paper's last paragraph before Theorem 4.10 ("a complete set of rewrite rules for $\langle\omega\rangle$ is easily found: we only need $\omega^d = 1$"), with the one non-trivial correction computed and checked.

7 Discussion

7.1 What is verified, and what is not

Everything stated as a theorem in §§2–6 is checked by Agda 2.8 with the standard library 2.4, under `-safe`, with no postulates and no holes; the projective chain is `-cubical-compatible` as well. The headline theorems — `unique-nf`, `simplified-presentation`, `v1-presentation`, `clifford-presentation` — are restated with full types in the index module `MainTheorems.agda`, which reaches, through them, everything those proofs rest on; the exact presentation `presentation-exact` is checked from its own root. Every theorem is parametric in the odd prime p, the width n, and a primitive root g.

Three things are *not* formalised, and it is worth being precise about them, because they delimit what "a machine-checked proof of the qupit paper's theorem" means here.

The glue facts. Our semantic groups are constructed: $\mathrm{Sp}(2n, \mathbb{Z}_p)$ is the group of linear symplectic bijections of $\mathbb{Z}_p^{2n}$, $\overline{\mathcal{P}}_n$ is $\mathbb{Z}_p^{2n}$ under addition, `Pauli⋊Sp` is their semidirect product, and the Clifford group with scalars is a central extension of that by a cocycle. That these are isomorphic to the groups of unitary matrices generated by the paper's H, S, CZ modulo (or not modulo) phases is classical — the paper's Proposition 2.25 and Theorem 2.26, citing Weil — and is not formalised. No complex number, matrix or unitary group appears in the qupit development; the matrix conventions appear only as a comment recording why exactly one rule picks up a phase. A verified matrix semantics would close this gap, and is the natural next step (§9).

The identification of the two central extensions. The structural Clifford group (Weyl cocycle on the natural semidirect product) and the presented one (a cocycle generated from generator data on the word groups) have quotients that are provably isomorphic — that is Theorem 2.1 — and cocycles that agree by construction, since the generator data is the Weyl cocycle pulled back along the presentation isomorphism; that the two *total* groups are isomorphic is not stated as a theorem.

The primitive root. Every module takes a generator g of $\mathbb{Z}_p^*$ and the witness `g-gen` as parameters. The existence of a primitive root modulo every prime is classical number theory and is not in the library; the only instance discharged is $p = 2$. This mirrors the paper, whose Figure 1 is stated for a chosen g.

7.2 Statistics

Table 3 gives the shape of the development. The reusable framework is about 15 000 lines; the qupit-specific development is about 64 000, of which the symplectic factor alone is 35 000 — confirming that the symplectic group is where the real work is — and the composition step 28 000. For comparison, the paper's own appendices and supplement are roughly a hundred pages. Two directories in the tree are not on the chain: `Simplified-V2`, a fifth rule set with the Pauli parts of every rule pushed to the right, which transports the theorem one step further along an identity isomorphism and which nothing imports; and the qubit Clifford developments, which are work in progress (§9).

Table 3. Size of the development (lines of Agda incl. comments, `wc -l`, September 2026).

Component	Files	Lines
Framework (`Word`, `Presentation`, `Circuit`, `Normalization`, `ForStdlib`)	54	14 848
of which presentation constructions (products, extensions, cocycles)	8	5 087
of which $\mathbb{Z}_p$ arithmetic, Fermat, primitive roots	8	1 861
Symplectic group	118	35 399
normalisation (boxes, coset table, tower, uniqueness)	35	8 352
box relations and supporting calculus (`BR`, `Lemmas`)	54	18 161
soundness, surjectivity, syntax, simplified rules	29	8 886
Projective Pauli group	3	1 220
Projective Clifford: semidirect, plumbing, shared calculus	26	17 996
of which `Simplified-V1` / `Paper-V0` / `Paper-V1`	22	16 133
Clifford with scalars	11	5 029
Other examples in the tree (symmetric, cyclic, wreath, Clifford+T, qubit Clifford)	107	27 552
Root index and notations	2	212
Not on the chain (`Simplified-V2`)	8	4 132
Total	329	106 388

Checking the exact presentation, with the interfaces of the projective chain cached, takes about six minutes on a laptop. Individual files have their own stories: the preamble of one two-qupit box-relation module costs nine CPU seconds merely to instantiate its rewriting tactics, so it is factored out and imported; the double inversion in M_x = `SHS'` x^{-1} x computes a Bézout witness twice and "makes even the equation `XM` ≡ `M`$^{-1}$ take minutes", so the multiplier is written over the primitive spelling; and the identification of the transported semidirect action with `ap` is `abstract` for the reason recorded in §6.2.

7.3 Proof-engineering lessons

Prove well-definedness by semantics. The single most consequential decision in the symplectic development is that the coset table's respect for the axioms — the paper's "the dirty gates get removed in the same way whichever rewrite one applies" — is proved not by analysing the table but by uniqueness one level down: two congruent inputs produce cosets with the same bottom-wire action, so by head injectivity the same coset; the residuals then agree by cancellation, given injectivity of $\uparrow$, which is faithfulness one wire below. This turns the coset tower into a well-founded induction on the width and removes any need to reason about rewrite paths. It is also the reason the tower does not close by evaluation as the symmetric group's does, and why §5.5 reads differently from the warm-up.

Transport, do not re-prove. Every rule set above the semidirect level is connected to the one below by an isomorphism of presentations, and theorems are carried across by `v1⇒pap`-style congruence lemmas. The swap calculus is proved once, in `Simplified-V1`, by transport from the symplectic development; `Paper-V1` carries only what its own isomorphism needs (its inherited *Ex*-conjugation and three-wire layers were deleted once transport made them redundant). The cost of this design is that `Grouplike` witnesses sometimes have to be proved natively rather than transported, because the chains below need right-cancellation and deriving it from a transported witness would be circular.

State calculi over an abstract relation. `Shared.PauliBase` proves the Pauli calculus and the *R*/*S*-spelling bridge once, over an arbitrary relation Γ that satisfies a handful of named hypotheses, and instantiates it at two rule sets; the one hypothesis whose proof differs between them, $XZ \approx ZX$, is a parameter. This is the Agda counterpart of the paper's practice of proving derived relations "from the complete rule set" in its supplement, with the dependency made explicit.

Make the semantics compute. Modelling $\mathrm{Sp}(2n, \mathbb{Z}_p)$ by transformations rather than matrices, with extensional equality of the action, made associativity and units definitional and let the ring solver discharge every generator obligation. It also made the semidirect action *definable* from the same phase space the Pauli presentation targets, which is what allowed the two well-definedness conditions of the semidirect theorem to be proved by soundness and completeness rather than by hand.

7.4 On the use of a language model

Claude was used to help write the Agda code of this development and to draft this paper from an outline written by the authors; the formalisation was designed, checked and corrected by the authors, and the paper was checked and revised by them. We record this both for transparency and because it bears on the argument of the paper: a verified development is one whose correctness does not depend on who, or what, wrote it. The typechecker's verdict does not depend on it.

8 Related Work

Generators and relations for quantum gate groups. The programme of presenting quantum gate groups by generators and relations, with completeness proved through normal forms, was opened by Selinger's presentation of the n-qubit Clifford group [Selinger 2015], whose staircase normal form and "push a generator through a box" rewrite system are the direct ancestors of the symplectic normal form verified here. The line continued with presentations of $U_n(\mathbb{Z}[\frac{1}{2}, i])$ by Bian and

Selinger [Bian and Selinger 2021] (building on Greylyn's presentation of $U_4(\mathbb{Z}[1/\sqrt{2}, i])$ [Greylyn 2014]), of the orthogonal groups $O_n(\mathbb{Z}[\frac{1}{2}])$ by Li, Ross, and Selinger [Li et al. 2021], of the real stabilizer operators by Makary, Ross, and Selinger [Makary et al. 2021], of the 2-qubit Clifford+T and 3-qubit Clifford+CS groups by Bian and Selinger [Bian 2023; Bian and Selinger 2023a,b], of the CNOT-dihedral operators by Amy, Chen, and Ross [Amy et al. 2018], and of 3-qubit Toffoli–Hadamard circuits by Amy, Ross, and Wesley [Amy et al. 2024]. The first such result in an odd prime dimension is the qutrit Clifford rule set of Li, Mosca, Ross, van de Wetering, and Zhao [Li et al. 2025], and its generalisation to all odd primes is the paper this work verifies [Bian et al. 2026a]. In every one of these papers the completeness proof is a normal-form argument checked by hand, and the supporting derivations occupy tens of pages; the Bian–Selinger papers are the exception in that their *syntactic* relation-simplification steps were checked in Agda. We have discussed the relation between the paper's proof and ours in §4.

Complete equational theories for circuits. A parallel programme, phrased in terms of PROPs and string diagrams rather than groups, seeks complete equational theories for quantum circuits. Clément, Heurtel, Mansfield, Perdrix, and Valiron gave the first complete equational theory for arbitrary quantum circuits [Clément et al. 2023]; Clément, Delorme, Perdrix, and Vilmart simplified it and extended it to circuits with ancillae and discarding [Clément et al. 2024b]; Clément, Delorme, and Perdrix made it minimal [Clément et al. 2024a]; and Clément's complete theory for Real-Clifford+CH circuits — the first for a finitely generated universal fragment without parameters or ancillae — comes with an appendix of 264 pages of equational derivations [Clément 2026]. Blake's recent work treats many near-Clifford fragments uniformly as PROPs, transferring completeness results and obtaining simpler and often minimal rule sets for qubit Clifford, real Clifford, Clifford+T (≤ 2 qubits), Clifford+CS (≤ 3 qubits), CNOT-dihedral and qutrit Clifford circuits [Blake 2026c]; gives a finite, dimension-uniform complete presentation of qudit circuits via polycontrolled PROPs [Blake 2026a]; and proves completeness for prime-dimensional phase-affine circuits through unique normal forms [Blake 2026b]. All of these are pen-and-paper. Behind circuit presentations of this kind stands Lafont's algebraic theory of Boolean circuits [Lafont 2003], which presented the monoidal categories of reversible and of quantum Boolean circuits by generators and relations and introduced the normal forms for linear and affine permutation circuits that the circuit-level normal forms above generalise. On the graphical side, the ZX-calculus has complete axiomatisations for the qubit stabilizer fragment, for Clifford+T, and — most relevant here — for the stabilizer fragment in odd prime dimensions [Backens 2014; Booth and Carette 2022; Jeandel et al. 2018; Poór et al. 2023]; as the paper we verify points out, completeness of a graphical calculus for *linear maps* does not yield a complete calculus for *unitary circuits*, which is why the circuit rule set needs its own proof.

Mechanised completeness. To our knowledge, machine-checked completeness proofs for quantum circuit rule sets are rare. Bian and Selinger's Agda developments [Bian and Selinger 2023a,b] verify that two *syntactic* presentations are equivalent and never mention a semantics. Choudhury, Karwowski, and Sabry mechanised, in HoTT-Agda, the presentation of the classical reversible language Π as a free symmetric rig groupoid [Choudhury et al. 2022], and Carette, Heunen, Kaarsgaard, and Sabry partially mechanised their square-root extension, including the complete completeness proof for the two-qubit Clifford fragment [Carette et al. 2024]; both are categorical coherence arguments rather than presentations of concrete groups. The Agda library this paper builds on [Bian and Feng 2026a,b] was designed to close exactly this gap — completeness of concrete relation sets against independently constructed groups — and its earlier case studies (symmetric groups, wreath products, Pauli groups, and the single-qubit and single-qutrit Clifford+T monoids as amalgamated products) are its first validation; the present paper is its first application to a research-level completeness theorem. Verified quantum compilers and program logics — QWIRE [Paykin et al. 2017], SQIR/VOQC [Hietala et al. 2021], VyZX [Lehmann et al. 2025], CoqQ [Zhou et al. 2023], Qbricks [Chareton et al. 2021] — prove the *soundness* of rewrites against a matrix semantics, never the completeness of a rule set; connecting our presented groups to such a semantics is discussed in §9.

Group theory in proof assistants. Lean's mathlib defines presented groups as quotients of free groups and builds Coxeter groups on them [The mathlib Community 2020, 2026]; Isabelle's Archive of Formal Proofs has free groups and the Nielsen–Schreier theorem [Breitner 2010; Kharim et al. 2023]; the Coq/MathComp school culminating in the Odd Order theorem [Gonthier et al. 2013] develops deep finite group theory over concrete permutation groups. None of these provides what this paper needs: verified coset enumeration for a *given* finite relation set against a *given* concrete group, packaged so that the certificate (a coset table, a section, five equations per level) is separated from the search that found it. The classical algorithms are described by Sims [Sims 1994] and in the Handbook of Computational Group Theory [Holt et al. 2005], whose Proposition 2.55 — a presentation of a group from presentations of a normal subgroup and the quotient — is the pen-and-paper form of the extension theorems we use.

9 Conclusion and Future Work

We have given a machine-checked proof that the qupit paper's rule set presents the multi-qupit Clifford group in every odd prime dimension: modulo scalars, the paper's Figure 1 presents $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$, and with the scalar ω and its single correction it presents a central extension of that group by $\langle \omega \rangle$; along the way, the simplified symplectic rules present $\mathrm{Sp}(2n, \mathbb{Z}_p)$, and the symplectic normal form — refined into a doubly inductive datatype of boxes and rows — is unique. All relations involve at most three qupits, and every result is parametric in p, n and a primitive root. The proof does not follow the paper: it factors the group, presents the factors, and composes the presentations with generic verified theorems, and it connects back to the paper's route by a verified isomorphism of presentations. We have argued that this organisation is not merely convenient for Agda but is the right way to structure such proofs: it confines the rewriting to the symplectic factor, where every rule is an honest equation, and it replaces the paper's hand-performed Pauli bookkeeping by one theorem proved for arbitrary semidirect products.

Future work. The immediate target is the remaining glue: a verified matrix semantics for the qupit gates, against which the constructed groups of §7 would be shown isomorphic to the groups of unitaries, closing the chain from Figure 1 to $U_{p^n}(\mathbb{Z}[1/p, \omega])$ end to end. Beyond that, we intend to verify the other presentations by generators and relations of quantum gate groups in the literature (§8). The qubit Clifford group [Selinger 2015] is under way: the tree already contains the qubit analogue of the projective chain, with its own semidirect and extension presentations, and the two-qubit Clifford+T and three-qubit Clifford+CS presentations [Bian and Selinger 2023a,b], whose syntactic halves were checked in Agda by their authors, are natural next instances for the amalgamation machinery of the library. The qutrit rule set [Li et al. 2025] is a special case of what is proved here at $p = 3$, though with its own spelling of the rules, and its verification is a plumbing step. Finally, the presentation-level constructions used here — semidirect products, central extensions from cocycles, and the extension recipe of Proposition 2.55 — and the supporting algebra (semidirect products, extensions, cocycles, and $\mathbb{Z}_p$ arithmetic with Fermat's theorem) are being prepared for submission to the Agda standard library.

Acknowledgments

We thank Neil J. Ross and Yuming Zhao, our coauthors on the rule set this paper verifies, for discussions throughout. We have used Claude to help with coding this project in Agda. We have used Claude to write this paper according to our outline; the paper was later checked and revised by us.

References

Matthew Amy, Jianxin Chen, and Neil J. Ross. 2018. A Finite Presentation of CNOT-Dihedral Operators. In *Proceedings of the 14th International Conference on Quantum Physics and Logic (QPL 2017) (Electronic Proceedings in Theoretical Computer Science, Vol. 266)*. 84–97.

Matthew Amy, Neil J. Ross, and Scott Wesley. 2024. A Sound and Complete Equational Theory for 3-Qubit Toffoli-Hadamard Circuits. In *Proceedings of the 21st International Conference on Quantum Physics and Logic (QPL 2024) (Electronic Proceedings in Theoretical Computer Science, Vol. 406)*. 1–43. arXiv:2407.11152.

Miriam Backens. 2014. The ZX-calculus is complete for stabilizer quantum mechanics. *New Journal of Physics* 16 (2014), 093021. arXiv:1307.7025.

Xiaoning Bian. 2023. *Generators and Relations for Some Classes of Quantum Circuits*. Ph. D. Dissertation. Dalhousie University.

Xiaoning Bian and Yuan Feng. 2026a. An Agda library for normal forms and complete relations for circuits. https://github.com/onestruggler/qupit.

Xiaoning Bian and Yuan Feng. 2026b. Normal Forms and Complete Relations for Circuits in Agda. Under submission.

Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, and Yuming Zhao. 2026a. A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions. Manuscript, July 2026. Agda code at https://github.com/onestruggler/qupit.

Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, and Yuming Zhao. 2026b. A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions: Derived Relations and Supplemental Proofs. Supplement to [Bian et al. 2026a], July 2026.

Xiaoning Bian and Peter Selinger. 2021. Generators and Relations for $U_n(\mathbb{Z}[1/2, i])$. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 145–164. arXiv:2105.14047.

Xiaoning Bian and Peter Selinger. 2023a. Generators and Relations for 2-Qubit Clifford+T Operators. In *Proceedings of the 19th International Conference on Quantum Physics and Logic (QPL 2022) (Electronic Proceedings in Theoretical Computer Science, Vol. 394)*. 13–28. arXiv:2204.02217.

Xiaoning Bian and Peter Selinger. 2023b. Generators and Relations for 3-Qubit Clifford+CS Operators. In *Proceedings of the 20th International Conference on Quantum Physics and Logic (QPL 2023) (Electronic Proceedings in Theoretical Computer Science, Vol. 384)*. 114–126. arXiv:2306.08530.

Colin Blake. 2026a. A Complete Equational Presentation of Qudit Circuits via Polycontrolled PROPs. In *51st International Symposium on Mathematical Foundations of Computer Science (MFCS 2026) (LIPIcs)*. arXiv:2602.09873.

Colin Blake. 2026b. Completeness for Prime-Dimensional Phase-Affine Circuits. arXiv:2603.06466.

Colin Blake. 2026c. Simpler Presentations for Many Fragments of Quantum Circuits. In *11th International Conference on Formal Structures for Computation and Deduction (FSCD 2026) (LIPIcs)*. To appear. arXiv:2602.09874.

Robert I. Booth and Titouan Carette. 2022. Complete ZX-Calculi for the Stabiliser Fragment in Odd Prime Dimensions. In *47th International Symposium on Mathematical Foundations of Computer Science (MFCS 2022) (LIPIcs, Vol. 241)*. 24:1–24:15. arXiv:2204.12531.

Joachim Breitner. 2010. Free Groups. Archive of Formal Proofs, https://www.isa-afp.org/entries/Free-Groups.html.

Jacques Carette, Chris Heunen, Robin Kaarsgaard, and Amr Sabry. 2024. With a Few Square Roots, Quantum Computing Is as Easy as Π. *Proceedings of the ACM on Programming Languages* 8, POPL (2024). arXiv:2310.14056.

Christophe Chareton, Sébastien Bardin, François Bobot, Valentin Perrelle, and Benoît Valiron. 2021. An Automated Deductive Verification Framework for Circuit-Building Quantum Programs. In *Programming Languages and Systems: 30th European Symposium on Programming (ESOP 2021) (Lecture Notes in Computer Science, Vol. 12648)*. 148–177.

Vikraman Choudhury, Jacek Karwowski, and Amr Sabry. 2022. Symmetries in Reversible Programming: From Symmetric Rig Groupoids to Reversible Programming Languages. *Proceedings of the ACM on Programming Languages* 6, POPL (2022), 6:1–6:32.

Alexandre Clément. 2026. A Complete Equational Theory for Real-Clifford+CH Quantum Circuits. arXiv:2602.06644. Main text 13 pages, appendix of 264 pages of equational derivations.

Alexandre Clément, Noé Delorme, and Simon Perdrix. 2024a. Minimal Equational Theories for Quantum Circuits. In *Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2024)*. arXiv:2311.07476.

Alexandre Clément, Noé Delorme, Simon Perdrix, and Renaud Vilmart. 2024b. Quantum Circuit Completeness: Extensions and Simplifications. In *32nd EACSL Annual Conference on Computer Science Logic (CSL 2024) (LIPIcs, Vol. 288)*. 20:1–20:23. doi:10.4230/LIPIcs.CSL.2024.20

Alexandre Clément, Nicolas Heurtel, Shane Mansfield, Simon Perdrix, and Benoît Valiron. 2023. A Complete Equational Theory for Quantum Circuits. In *Proceedings of the 38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2023)*. arXiv:2206.10577.

Georges Gonthier, Andrea Asperti, Jeremy Avigad, Yves Bertot, Cyril Cohen, François Garillot, Stéphane Le Roux, Assia Mahboubi, Russell O'Connor, Sidi Ould Biha, Ioana Pasca, Laurence Rideau, Alexey Solovyev, Enrico Tassi, and Laurent Théry. 2013. A Machine-Checked Proof of the Odd Order Theorem. In *Interactive Theorem Proving (ITP 2013) (Lecture Notes in Computer Science, Vol. 7998)*. 163–179.

Seth E. M. Greylyn. 2014. Generators and relations for the group $U_4(\mathbb{Z}[1/\sqrt{2}, i])$. MSc thesis, Dalhousie University. arXiv:1408.6204.

Kesha Hietala, Robert Rand, Shih-Han Hung, Xiaodi Wu, and Michael Hicks. 2021. A Verified Optimizer for Quantum Circuits. *Proceedings of the ACM on Programming Languages* 5, POPL (2021), 37:1–37:29.

Derek F. Holt, Bettina Eick, and Eamonn A. O'Brien. 2005. *Handbook of Computational Group Theory*. Chapman & Hall/CRC.

Emmanuel Jeandel, Simon Perdrix, and Renaud Vilmart. 2018. A Complete Axiomatisation of the ZX-Calculus for Clifford+T Quantum Mechanics. In *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2018)*.

Aabid Seeyal Abdul Kharim, Turga Venkata Hanumantha Prathamesh, Shweta Rajiv, and Rishi Vyas. 2023. Formalizing Free Groups in Isabelle/HOL: The Nielsen-Schreier Theorem and the Conjugacy Problem. In *Intelligent Computer Mathematics (CICM 2023) (Lecture Notes in Computer Science, Vol. 14101)*.

Yves Lafont. 2003. Towards an algebraic theory of Boolean circuits. *Journal of Pure and Applied Algebra* 184, 2–3 (2003), 257–310.

Adrian Lehmann, Ben Caldwell, Bhakti Shah, and Robert Rand. 2025. VyZX: Formal Verification of a Graphical Quantum Language. *ACM Transactions on Programming Languages and Systems* (2025). arXiv:2311.11571.

Sarah Meng Li, Michele Mosca, Neil J. Ross, John van de Wetering, and Yuming Zhao. 2025. A Complete and Natural Rule Set for Multi-Qutrit Clifford Circuits. In *Proceedings of the 22nd International Conference on Quantum Physics and Logic (QPL 2025) (Electronic Proceedings in Theoretical Computer Science, Vol. 426)*. 23–78. arXiv:2508.14670.

Sarah Meng Li, Neil J. Ross, and Peter Selinger. 2021. Generators and Relations for the Group $O_n(\mathbb{Z}[1/2])$. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 210–264. arXiv:2106.01175.

Justin Makary, Neil J. Ross, and Peter Selinger. 2021. Generators and Relations for Real Stabilizer Operators. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 14–36. arXiv:2109.05655.

Ulf Norell. 2007. *Towards a Practical Programming Language Based on Dependent Type Theory*. Ph. D. Dissertation. Chalmers University of Technology.

Jennifer Paykin, Robert Rand, and Steve Zdancewic. 2017. QWIRE: A Core Language for Quantum Circuits. In *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2017)*. 846–858.

Boldizsár Poór, Robert I. Booth, Titouan Carette, John van de Wetering, and Lia Yeh. 2023. The Qupit Stabiliser ZX-travaganza: Simplified Axioms, Normal Forms and Graph-Theoretic Simplification. In *Proceedings of the 20th International Conference on Quantum Physics and Logic (QPL 2023)*. arXiv:2306.05204.

Peter Selinger. 2015. Generators and relations for n-qubit Clifford operators. *Logical Methods in Computer Science* 11, 2:10 (2015). arXiv:1310.6813.

Charles C. Sims. 1994. *Computation with Finitely Presented Groups*. Encyclopedia of Mathematics and its Applications, Vol. 48. Cambridge University Press.

The Agda Community. 2026. The Agda standard library, version 2.3. https://github.com/agda/agda-stdlib.

The mathlib Community. 2020. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*. 367–381.

The mathlib Community. 2026. Mathlib documentation: PresentedGroup, CoxeterGroup, and the Nielsen-Schreier theorem. https://leanprover-community.github.io/mathlib4_docs/Mathlib/GroupTheory/FreeGroup/NielsenSchreier.html.

Li Zhou, Gilles Barthe, Pierre-Yves Strub, Junyi Liu, and Mingsheng Ying. 2023. CoqQ: Foundational Verification of Quantum Programs. *Proceedings of the ACM on Programming Languages* 7, POPL (2023), 29:1–29:33.